# A Survey on Decision Support in High-Mix Low-Volume Manufacturing

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**Abstract**

The increasing demand for highly customised products challenges the ability of manufacturing systems to sustain high levels of automation, particularly in high-mix low-volume (HMLV) environments. Although decision support has become a major research area in response to these challenges, existing reviews either focus narrowly on specific methods such as reinforcement learning for scheduling or address HMLV manufacturing without systematically analysing decision support across manufacturing levels. This paper addresses that gap through a survey of 205 publications on decision support in HMLV manufacturing.

The survey provides a structured synthesis of how decision support approaches are applied across four hierarchical manufacturing levels: process, line and cell, factory, and enterprise and supply chain. The reviewed studies are analysed through four research questions addressing: (1) the key drivers motivating decision support in HMLV manufacturing, (2) the optimisation objectives pursued, (3) the technologies and methods employed, and (4) the evaluation practices and limitations reported.

The analysis reveals a rapid growth in research activity over the past decade, with a strong methodological concentration around scheduling problems, which account for 69.8% of the reviewed literature, while areas such as logistics and supply chain (6.3%) and quality and maintenance (2.9%) remain comparatively underexplored. Reinforcement learning, graph neural networks, digital twins, and metaheuristic optimisation emerge as dominant technologies and methods, within dynamic scheduling contexts.

The findings show that current research is concentrated on simulation-based scheduling optimisation, with comparatively limited attention to cross-level decision support, industrial validation, human-centred decision-making, and integrated decision-support architectures spanning multiple manufacturing levels. Based on these findings, the survey identifies key research gaps and outlines future directions for decision support in HMLV manufacturing.

These findings should be considered alongside the search and screening process, which relies primarily on broad HMLV, manufacturing, and decision-support terminology and may therefore under-represent research using specialised, industry-specific terminology.

# 1 Introduction

Manufacturing is increasingly shifting from high-volume, low-mix production towards High-Mix Low-Volume (HMLV) production, driven by increasing demand for product customisation and shorter product life cycles [1–3]. In HMLV environments, manufacturers produce a wide variety of products in small batches, often in a Make-To-Order (MTO) or Engineer-To-Order (ETO) setting [2], as these environments are characterised by high variability in both product and process requirements. This type of production is common in industries such as semiconductor manufacturing [4], aerospace [5], electronics, metal and machining, and contract manufacturing, where each order may require distinct processing routes, tooling, and setup configurations [1].

The characteristics of HMLV production create significant challenges for operational decision-making. Frequent changeovers, variable processing times, non-standard product routings, and unpredictable demand patterns make it difficult to plan and control production using traditional methods [6]. Decisions must be made at multiple levels of the manufacturing system: from individual machine parameters and process settings, through the coordination of robotic cells and assembly lines, to factory-wide scheduling, and enterprise- and supply chain-level planning [7]. At each level, the high product variety introduces complexity that is less pronounced in mass production, and the low volumes mean that the cost of poor decisions is spread across fewer units, making each decision more consequential [8]. In this work, we consider four hierarchical levels: Process Level (L0), Line and Cell Level (L1), Factory Level (L2), and Enterprise and Supply Chain Level (L3). This is illustrated in Figure 1. These four levels are inspired by the ISA-95 / IEC 62264 standard [9, 10]. However, they differ slightly, as the Level 0 and Level 1 of the ISA-95 standard here have been merged into 1 level, level 0 (process level) as these are interconnected and hard to distinguish within the literature. How the ISA-95 / IEC 62264 levels align with our levels is shown in table 3. Other examples of distinguishing manufacturing within levels, includes the RAMI 4.0, which builds on the ISA-95 standard by adding more dimensions to the classification [11].

The emergence of Industry 4.0 technologies has opened new possibilities for addressing these challenges. Advances in Reinforcement Learning (RL), Graph Neural Networks (GNNs), Digital Twins (DTs), Internet of Things (IoT), and Cyber-Physical Systems (CPSs) provide the computational tools needed to build Decision Support Systems (DSSs) that can handle the dynamic and uncertain nature of HMLV production [12–14]. The rapid growth of research in this area, particularly in recent years, makes it timely to take stock of what has been achieved, what methods and technologies are being used, and where gaps remain.

Several reviews have examined parts of this landscape, covering either HMLV manufacturing broadly [1] or specific methods such as RL for job-shop scheduling [15, 16]. However, no existing review systematically examines decision support across all manufacturing levels in the HMLV context (see Section 2). This survey will address this gap. In this survey, decision support is used as a broad term encompassing methods and approaches that support decision-making. The term DSS is used more specifically when referring to an integrated system-level implementation.

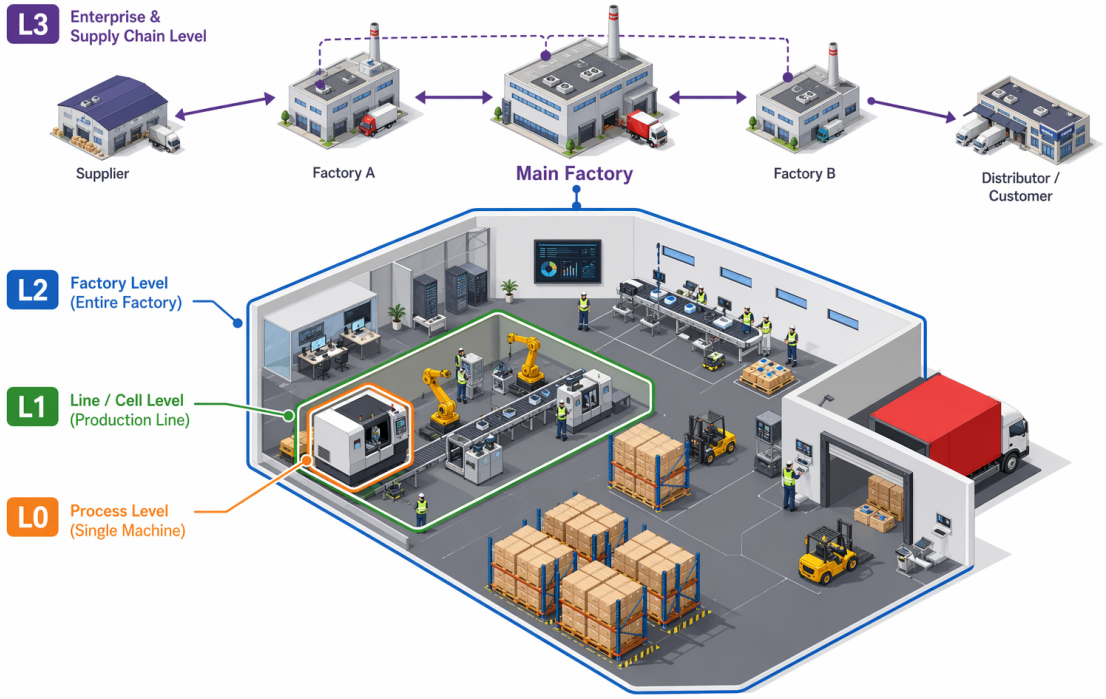



Figure 1: High-level overview of the manufacturing levels.

## 1.1 Research Objectives and Questions

The objective of this survey is to provide a structured synthesis of decision support approaches in HMLV manufacturing across four dimensions: the drivers behind existing approaches, the objectives they pursue, the technologies and methods they employ, and how their effects are evaluated. In doing so, the survey aims to offer an entry point for researchers who are new or interested in the field. Specifically, the contributions of this paper are fourfold: (i) to characterise decision support approaches and their underlying problem settings, (ii) to analyse how decision support is distributed across different manufacturing levels (L0 to L3), (iii) to identify the predominant technologies and methods employed, and (iv) to highlight limitations and research gaps in current approaches. With this in mind, the following Research Questions (RQs) guide the scope and structure of the survey:

- **RQ1: What drives the development and application of decision support approaches in HMLV manufacturing?**
  This RQ examines the common decision areas and problems addressed, including challenging characteristics such as dynamic environments, stochastic job arrivals, real-time requirements, and scalability, and how these drivers manifest across different production levels.
- **RQ2: What objectives do decision support approaches have in HMLV manufacturing, and how are their effects measured?**
  This RQ seeks to identify the decision support objectives in HMLV manufacturing, such as minimising makespan, minimising energy consumption, resource allocation efficiency, and robustness/resilience.

- **RQ3: Which technologies and methods are used to enable decision support in HMLV manufacturing?**
  This RQ examines which technologies and methods are most widely used and whether similarities or correlations exist between approaches applied at different production levels.
- **RQ4: How are the effects of decision support approaches for HMLV manufacturing evaluated, and what limitations are reported?**
  This RQ analyses the evaluation methodologies (e.g., simulation studies, case studies, experimental benchmarks, or industrial deployments). In addition, RQ4 identifies limitations related to applicability, scalability, robustness, and practical deployment in real manufacturing environments.

The remainder of this paper is structured as follows. Section 2 positions this survey against existing reviews. Section 3 describes the methodology used to select and analyse the relevant papers. Section 4 presents the quantitative results of the synthesis. Sections 5.1–5.4 provide a qualitative review of the included papers, structured by manufacturing level and RQ. Finally, Section 6 discusses the results and Section 7 concludes with implications for future research.

# 2 Related Literature

This section positions the present survey against existing reviews and identifies the gap it aims to fill. Table 1 summarises the most relevant review papers, grouped by their primary focus area. The selection is not exhaustive but aims to cover the main research streams that intersect with decision support in HMLV manufacturing.

## 2.1 Existing Reviews of HMLV Manufacturing

Two reviews address HMLV manufacturing broadly. Gan et al. [1] review 152 papers published between 2000 and 2022, covering production planning, scheduling, layout design, and Industry 4.0 technologies in HMLV contexts. Their findings show that production planning is the dominant research area, that the vast majority of the literature is not industry-specific, and that the semiconductor and electronics industries account for most of the validated work. They identify a shift from algorithm-based approaches towards Industry 4.0-enabled DSSs, noting that comprehensive DSSs covering multiple aspects of the value chain are a key direction for future research.

While this review is the closest to the present survey in scope, it does not systematically analyse decision support approaches across the four manufacturing levels (L0-L3) or along the four research questions of this survey: drivers, objectives, technologies and methods, and evaluation and limitations. Tahmina et al. [17] provide a shorter overview of smart manufacturing solutions for HMLV in defence and aerospace, covering system-level strategies such as group technology and lean practices alongside process level solutions such as robotics and additive manufacturing. Their review highlights data-driven decision-making as a future trend but does not systematically review decision support methods or DSS architectures.

## 2.2 Reviews of RL and AI for Scheduling

A substantial body of recent reviews examines RL and related methods for job-shop scheduling. Wang et al. [15] review 93 papers on RL for job-shop scheduling from 2016 to 2023, analysing single-agent and multi-agent approaches and identifying the curse of dimensionality, generalisation, and training time as key open challenges. Chen and Zhang [18] focus specifically on Deep Reinforcement Learning (DRL), reviewing 60 papers from 2019 to 2024 with detailed analysis of state feature design, action spaces, and neural network architectures. Dalle Mura and Dini [19] survey the use of GNNs for job-shop scheduling, covering architectures that combine GNNs with RL and multi-agent RL. Ngwu et al. [16] review AI-driven approaches for dynamic job-shop scheduling, assessing how evolutionary heuristics, machine learning, and RL handle real-time disruptions.

These reviews provide detailed methodological coverage but share a common limitation: they focus on scheduling methods in isolation, without situating them in the broader context of HMLV manufacturing or considering decision support needs beyond scheduling, such as process level optimisation, Human Robot Collaboration (HRC), or information provisioning.

## 2.3 Reviews of Decision Support in Manufacturing

At a broader level, Kasie et al. [13] survey DSSs in manufacturing and propose a theoretical framework that integrates AI, specifically case-based reasoning and rule-based reasoning, with Discrete-Event Simulation (DES) and database management. Their review traces the evolution of DSSs from the 1970s and identifies the integration of AI with simulation as the most promising direction for manufacturing decision support. However, this review predates the rise of DRL and GNNs for scheduling, and does not address HMLV specifically. In a more focused scope, Meilanitasari and Shin [20] review prediction and optimisation approaches for scheduling in job-shop flexible manufacturing systems, with attention to uncertainty factors such as variable processing times and machine breakdowns. Johansen et al. [8] review the role of automation in managing the complexities of HMLV production, focusing on physical automation challenges such as flexible assembly, HRC safety, and robot programming rather than computational decision support. At the process level, Böttjer et al. [21] review 96 papers on unit-level DTs in manufacturing, covering machine tools, additive manufacturing, and robotic applications. Their findings highlight a gap between laboratory-scale DT implementations and real-world deployment, as well as challenges in physical model fidelity and semantic interoperability. While not framed as decision support, many of the reviewed implementations use DTs for real-time control and parameter optimisation at the machine level, which overlaps with the process level decision support that this survey examines.

## 2.4 Gap and Contribution of This Survey

Table 1: Comparison of related reviews with the present survey. Manufacturing levels: L0 (process), L1 (line/cell), L2 (factory), L3 (enterprise/supply chain)

| Review | Year | Primary Focus | HMLV | DSS | Level | RQ1 | RQ2 | RQ3 | RQ4 |
|---|---|---|---|---|---|---|---|---|---|
| **HMLV manufacturing reviews** | | | | | | | | | |
| [1] | 2023 | HMLV production broadly | ✓ | ○ | L2/L3 | ○ | ✓ | ✓ | ○ |
| [17] | 2023 | Smart mfg. for HMLV (defense) | ✓ | | L0/L2 | ○ | ○ | ✓ | ○ |
| **RL and AI for scheduling reviews** | | | | | | | | | |
| [15] | 2025 | RL for job-shop scheduling | | ○ | L2 | | ○ | ✓ | ✓ |
| [16] | 2025 | AI for dynamic job-shop | | ○ | L2 | | | ✓ | ✓ |
| [18] | 2025 | DRL for job-shop scheduling | | ○ | L2 | | | ✓ | ○ |
| [19] | 2025 | GNN for job-shop scheduling | | ○ | L2 | | ○ | ✓ | ✓ |
| **DSS and domain-specific reviews** | | | | | | | | | |
| [13] | 2017 | DSS in manufacturing (AI+DES) | | ✓ | L0/L1/L2/L3 | ✓ | ○ | ✓ | ✓ |
| [20] | 2021 | Scheduling in FMS | | ○ | L1/L2 | ○ | ✓ | ✓ | ✓ |
| [8] | 2021 | Automation & HRC in HMLV | ✓ | | L1/L2 | ○ | | ○ | ✓ |
| [21] | 2023 | Unit-level DT in mfg. | | ○ | L0 | ○ | | ✓ | ✓ |
| **This survey** | **2026** | **DSS in HMLV mfg.** | ✓ | ✓ | L0/L1/L2/L3 | ✓ | ✓ | ✓ | ✓ |

✓ = explicitly and systematically addressed; ○ = partially or indirectly addressed; blank = not covered. The classification reflects the extent to which each review explicitly analyses or structures its discussion around the corresponding manufacturing levels and research questions addressed in this survey. Reviews were classified as partially addressed when a topic was discussed only as a secondary aspect or without a dedicated analytical focus.

The existing reviews cover either the HMLV manufacturing context without a systematic focus on decision support methods, or specific methods (primarily RL for scheduling) without grounding them in the HMLV context. Notably, Gan et al. [1] explicitly identify comprehensive DSSs as a needed direction for future HMLV research. More broadly, no review systematically examines decision support across all manufacturing levels, from individual machines (L0) through production lines (L1) and factories (L2) to enterprise and supply chain (L3), while simultaneously analysing what drives these approaches (RQ1), what objectives they pursue (RQ2), which technologies and methods they use (RQ3), and how they are evaluated (RQ4).

This survey addresses this gap by reviewing 205 papers identified through the Scopus-based bibliographic database search strategy through the lens of four RQs, structured across four manufacturing levels. In doing so, it connects the methodological depth of the scheduling-focused reviews with the manufacturing context of the HMLV reviews, while also covering decision support areas beyond scheduling that have received less attention in the literature. An interactive overview of all included papers is available at `https://hmlv-survey.vercel.app/`, where researchers and practitioners can filter and explore the corpus by manufacturing level, decision area, and technology. The complete reference list for the survey corpus, including papers not explicitly cited in this survey, is provided in the Supplementary Materials.

# 3 Methods

This survey is inspired by the systematic article collection process Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [22]. PRISMA provides a set of guidelines that help authors to create transparent, systematic reviews. This is particularly with respect to the identification, selection, assessment, and synthesis of relevant studies. PRISMA was first published in 2009, updated in 2021, updating the guide in reporting systematic reviews. This includes the identification of new studies via other methods beyond traditional database searches, such as websites, organisations and snowballing (i.e., reference and/or citation searching). Figure 2 shows a flow diagram of the study selection process used in this study, adapted from [22]. The coming sections describe the identification, screening and eligibility phases performed in this survey. In section 3.1, the literature search procedure is described. In sections 3.2, 3.3, and 3.4, screening and eligibility assessment of the papers are performed, including automated screening, manual screening, and snowballing.

## 3.1 Literature identification

Scopus was selected as the primary database for literature retrieval due to its broad, comprehensive coverage of academic fields, advanced search and filtering capabilities, comprehensive citation tracking tools, and support for exporting results in an appropriate format, enabling a systematic and reproducible search process [23]. Based on the purpose of the survey, a set of keywords has been identified. To account for variations in terminology related to HMLV and decision support, multiple keyword variants (including abbreviations and equivalent expressions with differing word order and hyphenation) were included to broaden the search and increase the coverage of relevant papers. The search keywords include three core keywords: *High-Mix Low-Volume*, *Manufacturing*,

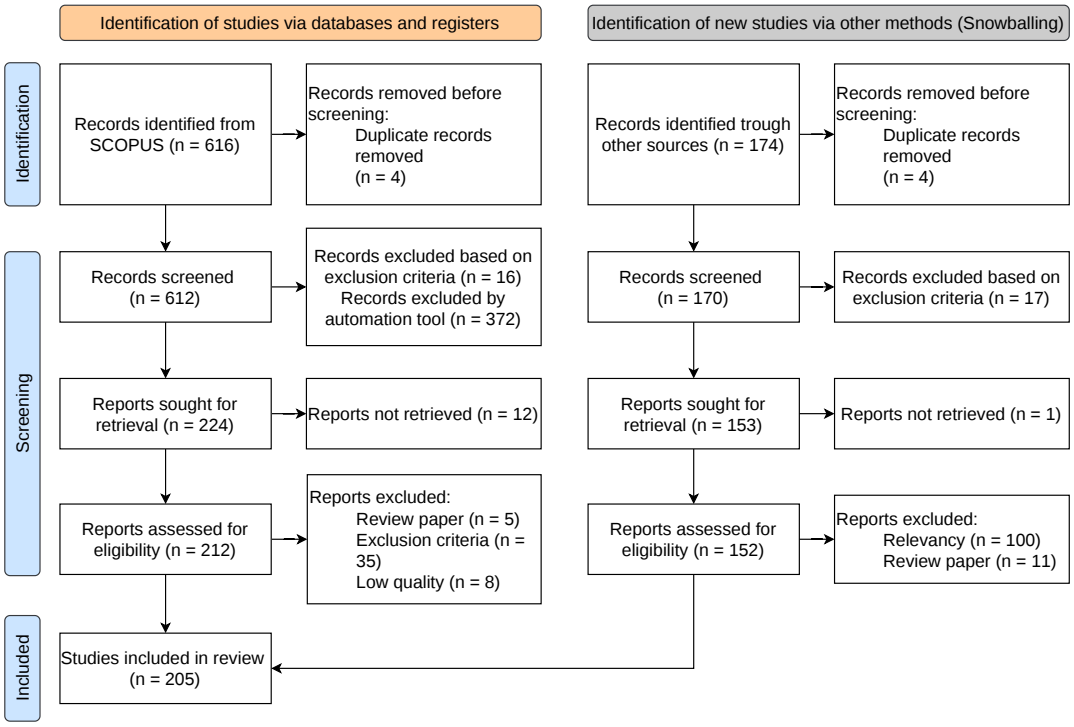



Figure 2: PRISMA 2020 flow diagram of the study selection process adapted from [22].

and *Decision Support*, with associated variants and extensions to capture differences in terminology. All keywords and their extensions are presented in Table 2

To narrow the number of papers, publications were restricted to the subject areas of *Engineering* and *Computer Science*, and to the publication years 2018-2025 (inclusive), to focus on recent and relevant research reflecting the recent development of modern Artificial Intelligence (AI). The search was further restricted to document types classified as conference papers or journal articles and to publications written in English.

This resulted in the search query on Figure 3, formulated in accordance with the Scopus advanced search syntax. Applying the search query identified 616 academic papers from journals and conferences. The search was performed on 5 January 2026.

```
(TITLE-ABS-KEY(hmlv OR "high mix low volume" OR "high-mix low-volume" OR
"low volume high mix" OR "high variety low volume" OR "make-to-order" OR
mto OR "small batch" OR "custom manufacturing" OR "mass customization" OR
"job shop" OR "configure-to-order" OR cto OR "short production runs" OR "engineer-to-order" OR "ETO")
AND TITLE-ABS-KEY("manufact*" OR "production" OR "industr*" OR "Industry 4.0")
AND TITLE-ABS-KEY("Decision Support" OR "Decision-support" OR
"Decision Support System" OR "DSS" OR "Decision Making")
AND PUBYEAR > 2017 AND PUBYEAR < 2026)
AND (LIMIT-TO(SUBJAREA,"ENGI") OR LIMIT-TO(SUBJAREA,"COMP"))
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp"))
AND (LIMIT-TO(LANGUAGE,"English"))
```

Figure 3: Scopus advanced search query.

As an introductory step to the screening phase, four meta-classes are made by the authors. To

Table 2: Search keyword structure

| **Core keyword** | **Extended keywords** | **Description** |
|---|---|---|
| High-Mix Low-Volume | hmlv; high mix low volume; low volume high mix; high variety low volume; make-to-order; MTO; small batch; custom manufacturing; mass customization; job shop; configure-to-order; CTO; short production runs; engineer-to-order; ETO | Manufacturing paradigms are characterised by high product variety and low production volumes. |
| Manufacturing | manufact*; production; industr*; Industry 4.0 | General industrial and manufacturing context papers. |
| Decision support | decision support; decision-support; decision support system; DSS; decision making | Methods and systems supporting or enabling decision-making processes. |

identify at which level of manufacturing the paper is concerned. In Table 3, each class is listed and described.

Table 3: Meta-classes, corresponding ISA-95 / IEC 62264 levels, and descriptions, used for classifying study scope

| **Tag** | **ISA-95/ IEC 62264 [10]** | **Meta-class** | **Description** | **Examples** |
|---|---|---|---|---|
| L0 | L0 - L1 | Process Level | Focus on a single machine, workstation, or manufacturing process. | Individual CNC machine, single robot, machining process |
| L1 | L2 | Line and Cell Level | A group of interacting machines or processes forming a production line or manufacturing cell. | Manufacturing cell with two robots and two CNC machines |
| L2 | L3 | Factory Level | An entire factory or plant consisting of multiple production lines or cells, or studies addressing a group of machines within one factory. | Complete production facility, multi-line factory |
| L3 | L4 | Enterprise and Supply Chain Level | Multiple factories within the same organisation, focusing on coordination, planning, or integration across sites, as well as interactions between multiple organisations, including external partners, across the supply chain. | Multi-site manufacturing enterprise or Supplier-manufacturer-distributor network. |

## 3.2 Automation tool

A relevance script was used to filter the papers and devise the *most relevant* papers, while preserving transparency and reproducibility. The code is available at GitHub repository[1]. The script analyses and evaluates each paper based on its textual content (title, abstract, and keywords) and assigns a relevance score according to predefined logic rules. The procedure consists of four main stages: text aggregation, keyword extraction, scoring logic, and classification into relevance levels, see Table 4. The detailed implementation, including keyword definitions and classification logic, is provided in section A.1.

Table 4: Relevance scoring scheme.

| **Score** | **Meaning** |
|---|---|
| ++ | Most relevant |
| + | Relevant |
| * | Possibly relevant/uncertain |
| – | Not relevant |

For each paper, the script records the assigned relevance label, a short reasoning message, and all detected technologies and methods. Where technologies and methods was predefined, and further elaborated in the individual reviews. Finally, the results are exported to an Excel file in a structured table format for easy filtering and manual validation. After running the scripts, 372 unique papers were filtered from the initial search. Of these, 36 were classified as possibly relevant (*) and 337 as not relevant (–). The 36 borderline cases that did not meet the inclusion threshold are presented at `https://hmlv-survey.vercel.app/` for interested readers. A validation of the script was performed by manually reviewing a random sample of 30 papers classified as not relevant. Of these, 2 were judged to be potentially relevant outside the scheduling field, indicating that the relevance script is not perfect and may exclude relevant papers. The source code is available on GitHub, and contributions aimed at improving the algorithm are welcome.

The remaining 240 papers, classified as relevant (+) or most relevant (++), were carried forward to the subsequent manual screening and eligibility assessment.

## 3.3 Manual Screening and Eligibility Criteria

After automated classification, the remaining papers were manually screened. This proceeded in three stages: (1) title and abstract screening to exclude non-relevant papers based on predefined inclusion and exclusion criteria (see Table 5), (2) retrieval of full texts, with unretrievable papers excluded, and (3) full-text eligibility assessment, with inclusion of relevant studies meeting inclusion criteria and exclusion of those meeting any of the exclusion criteria.

[1]Github repository: https://github.com/MartinBraas-AU/HMLV-survey

Table 5: Inclusion and exclusion criteria

| Inclusion criteria | Exclusion criteria |
|---|---|
| The paper targets a HMLV manufacturing context. | The paper has no relevance to the scope of this survey. |
| The paper falls within the defined manufacturing levels of interest. | The paper is purely conceptual. |
| The contribution enables or implements decision support. | The paper is not a conference paper or journal article. |
| | The paper is identified as a duplicate. |
| | The paper has a limited contribution or low quality[2]. |

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The screening process was conducted by the reviewing co-authors. In Table 6, the reviewing co-authors and their corresponding tags are listed.

Table 6: Reviewing Co-authors in the PRISMA process.

| ID | Reference | ID | Reference | ID | Reference |
|---|---|---|---|---|---|
| P1 | Braas, M. | P4 | Petersen, M.E.S. | P7 | Yousaf, S. |
| P2 | Birkebæk, M.T. | P5 | Lunding, R.S. | | |
| P3 | Dzigurski, F. | P6 | Fibiger, J. | | |

Papers were first grouped by manufacturing level and subsequently distributed among the reviewing co-authors. Allocation was performed manually with the aim of assigning a similar number of papers to each co-author while keeping assignments largely within a single manufacturing level. Each paper was assigned a reviewing co-author tag (see Table 6), enabling traceability of assignments. During the review phase, the co-authors met regularly to discuss ambiguous cases, align category definitions, and resolve screening and classification disagreements. Papers were therefore reviewed and validated collaboratively, although papers were not independently screened by multiple reviewers. The full mapping between papers and tags is provided in the master sheet available in the project's GitHub repository[3].

## 3.4 Identification through other sources

In addition to PRISMA, citation searching, referred to as snowballing, is used to identify relevant papers not included in the initially identified records from Scopus. The studies included were used as seed articles. Backward (reference list) and forward (citation) searches were performed using Scopus and Google Scholar on these seed articles. The manual procedures are repeated for these as well, including the exclusion/inclusion criteria. This is illustrated on Figure 2 in grey.

To identify generally well-connected papers, a script is used to identify those cited by the queried papers and those citing the queried papers. In doing this, rule-based filters were applied to identify papers for screening and eligibility review. Papers cited by the seed articles were included if they were cited at least 15 times within the set. Papers citing the seed articles were included if they had

[2]"Limited contribution" refers to studies without a clearly distinct contribution or extension compared with existing work. "Low quality" refers to insufficient methodological or experimental support, including inadequately described methods, missing experimental evaluation, or claims not sufficiently supported by the reported results to meet the inclusion criteria of this survey.

[3]GitHub repository: https://github.com/MartinBraas-AU/HMLV-survey

at least 5 total citations, were published between 2020 and 2026, and cited at least 7 of the seed articles. In total, 3788 citing papers and 5677 cited papers were identified. From these two sets, 26 papers (11 cited and 15 citing) were automatically sourced for manual screening and eligibility review.

Upon the review of the acquired papers, the number of papers in each level can be seen in the quantitative analysis (Figure 5). This shows a higher concentration of studies at the factory level decision support compared to other manufacturing levels. To strengthen the comprehensiveness of this Systematic Literature Review (SLR), manual identification is performed in the lesser represented levels: machine/process, line and cell, and enterprise and supply chain.

In summary, the study selection process combines automated filtering, manual screening, and snowballing in a sequential order. Initially, records are retrieved from Scopus and supplementary papers through snowballing. These records obtained from Scopus were first processed by a relevance script, which filtered out a portion of less relevant studies. The remaining papers were then subjected to manual screening based on title and abstract, followed by full-text eligibility assessment of accessible/retrievable papers

Afterwards, records identified through snowballing were directly subjected to the same manual screening and eligibility steps, without prior automated filtering. Across these pages, duplicates, inaccessible papers, and studies not meeting the inclusion criteria were excluded. The final set of papers (205) represents those that satisfy the combined screening and eligibility assessment. A high-level overview of this process, including the flow between stages and the distribution of included and excluded papers, is illustrated in Figure 4.

An overview of the included papers and extracted data is provided in `https://hmlv-survey.vercel.app/`. The interactive interface enables researchers and practitioners to explore the corpus of papers across manufacturing levels, with filtering by decision area, year, technologies and methods, and manufacturing level. Implementation details can be found on Github Repository[4].

## 3.5 Data Analysis

To analyse the records collected and included in the survey, a deductive and inductive approach is used. Predefined categories were initially created. These were: Manufacturing level, Methods, Technologies, Evaluation, Decision area, and Data source. The categories were made based on the purpose of the survey. Alongside the review process, additional categories were introduced to capture more relevant information about the papers. Papers spanning multiple manufacturing levels were assigned to the primary level addressed, while cross-level relationships were also identified. For each included paper, data were extracted into a shared master sheet using tags and supporting text. After the initial extraction, two reviewing co-authors reviewed all included papers and checked the extracted tags and text for consistency with the finalised categories.

## 3.6 Evaluation classes of decision support studies

A set of evaluation classes is defined to systematically categorise how decision support approaches are evaluated in the literature. This relates both to the environment in which the experimentation is performed and performance measured, but also to the data sources used in the evaluation. The

[4]Github repository: `https://github.com/MartinBraas-AU/HMLV-survey`

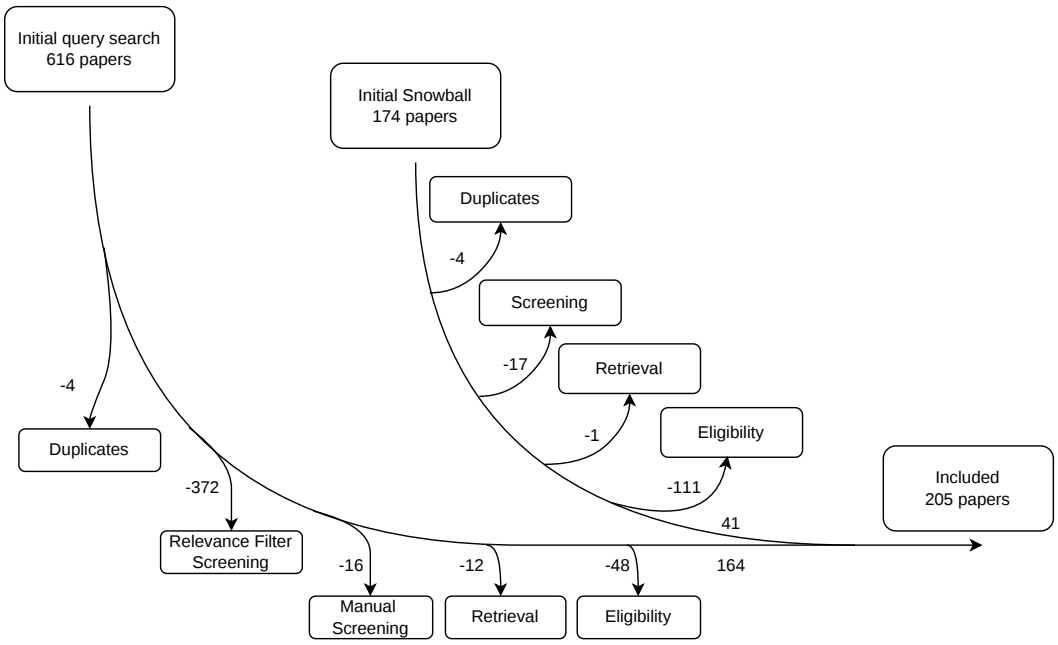



Figure 4: High-level overview of the paper selection process. Negative numbers indicate the number of records removed at each step. Positive numbers indicate records that survived all selection steps in the corresponding stream and were included in the final corpus.

classification is developed based on a preliminary screening of a subset of the reviewed studies and uses known terminology used in research.

The classes for the evaluation environment are as follows: Industrial case study evaluation, Laboratory case study evaluation, Simulation-based evaluation, and instance or static computational-based evaluation. Depending on the research field, a case study has different definitions and criteria for what can be seen as case study research. In this SLR, a case study research is shortly defined as an empirical research methodology intended to investigate a contemporary phenomenon within its real-life context [24]. To distinguish between the contexts for which the case study evaluation is performed, two subclasses named industrial case study and lab case study are used in this SLR. Below is each of the evaluation environment classes described:

- **Industrial setting**: Evaluation conducted in a real industrial or manufacturing environment, where the decision support approach is implemented or used in practice. The decision support approach's performance is measured in actual operations.
- **Laboratory setting**: Evaluation conducted as a controlled experiment in a physical test environment representing a specific system configuration. This could be a pilot manufacturing setup, mock-ups, experimental manufacturing machine, line or cell, or testbeds designed to replicate manufacturing conditions under controlled settings.
- **Simulation**: Evaluation performed using computational models that represent system behaviour. These models, both deterministic and stochastic, serve as the experimental environment, such as a manufacturing or decision context. Evaluation is then conducted by

executing the model. The understanding of simulation modelling follows [25].

- **Static evaluation**: Evaluation based on solving predefined problem instances without modelling or executing system dynamics, and no feedback to the simulated environment. This includes computing solutions to fixed datasets or optimisation problems. No system behaviour evolves during evaluation. This kind of evaluation environment is often used in metaheuristics or simple machine learning static inference, prediction and classification [26, 27].

In addition to the evaluation environment classes, the origin of the data used in experiments is classified. How data is obtained influences the realism, reproducibility, and generalisability of decision support approaches.

- **Industrial data**: Empirical data obtained from real manufacturing systems or operational environments.
- **Laboratory data**: Data collected from controlled experimental setups, testbeds, or pilot-scale manufacturing systems designed to replicate production conditions in a laboratory environment.
- **Synthetic data**: Data created through stochastic or procedural generation, including data generated by simulation models.
- **Benchmark data**: Publicly available or widely reused standard datasets designed to support reproducible comparison across multiple studies.

# 4 Quantitative analysis

This section presents the quantitative characteristics of the 205 reviewed papers.

Interactive versions of all figures and the full underlying dataset are available at `https://hmlv-survey.vercel.app/`. Furthermore, the distributions of industry domains, technologies, geographical origin, evaluation settings, and data sources are available at `https://hmlv-survey.vercel.app/dashboard`.

## 4.1 Research Interest Across Manufacturing Levels

Figure 5 shows the distribution of publications by year, stacked by manufacturing level. Research interest in decision support for HMLV manufacturing has grown substantially, rising from a single publication in 2016 to 81 in 2025, with a slight drop in 2023. This growth is mostly contributed by L2 and L3, whereas L1 has seen slower growth and L0 has remained relatively constant. The L2 accounts for 73.7% of the corpus, while L0 (14 papers), L1 (13 papers), and L3 (27 papers) each represent a much smaller share. Note that the 2026 count (3 papers) reflects only a partial year.

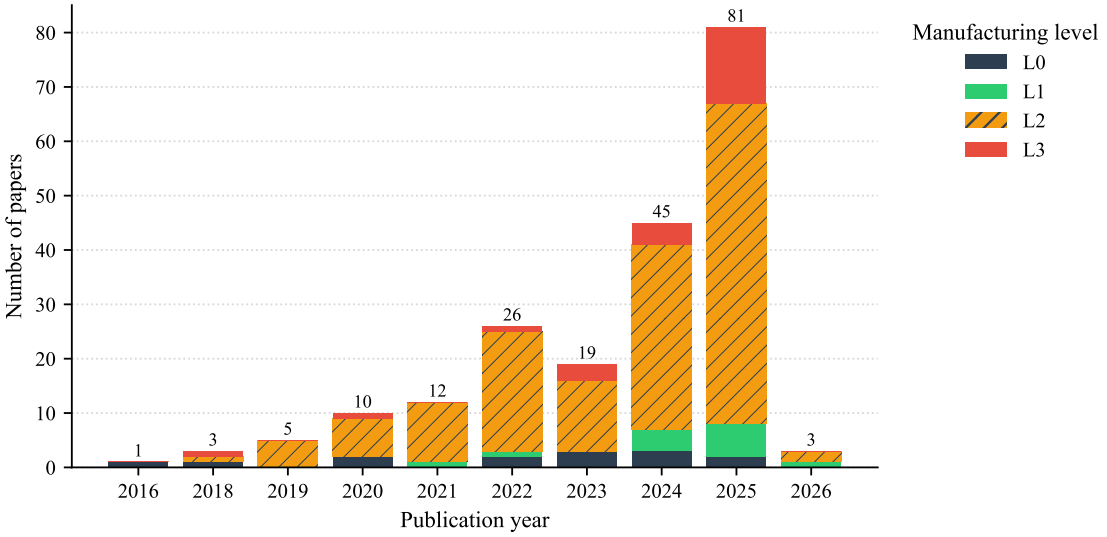



Figure 5: Number of publications per year. Where each bar is color coded based on manufacturing level. This indicated an increasing interest from the research community within decision support for HMLV manufacturing. The papers outside the 2018-2025 search window originate from snowballing.

## 4.2 Decision Support Focus

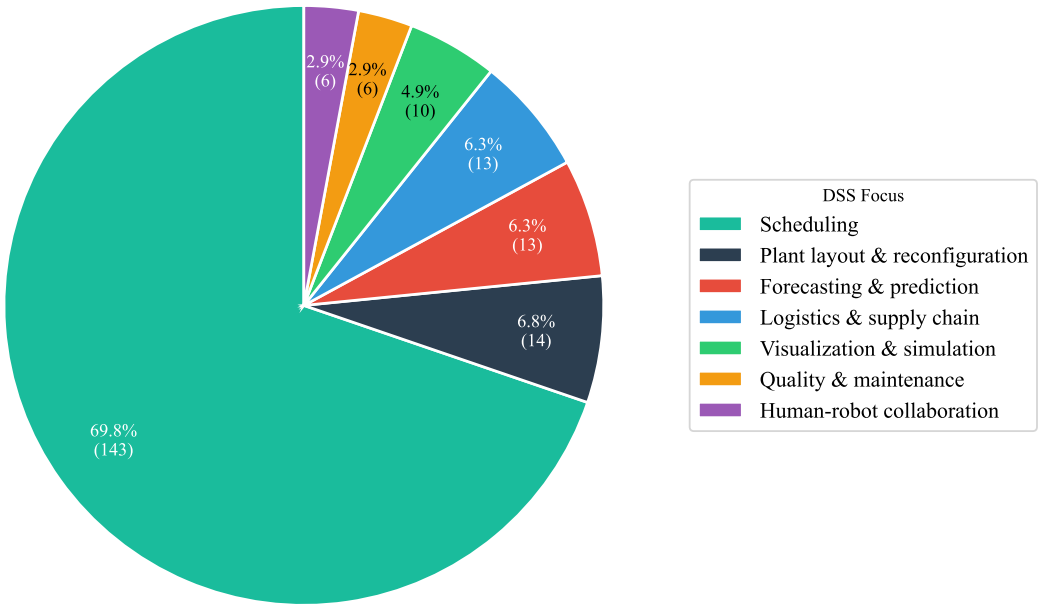



Figure 6: Reviewed papers distributed into categories based on their DSS focus application.

Figure 6 shows the distribution of papers by decision support focus area. Scheduling accounts for 69.8% of the reviewed literature. The remaining papers are distributed across logistics and supply chain, plant layout and reconfiguration, forecasting and prediction, visualisation and simulation, quality and maintenance, and HRC.

## 4.3 Industry Distribution

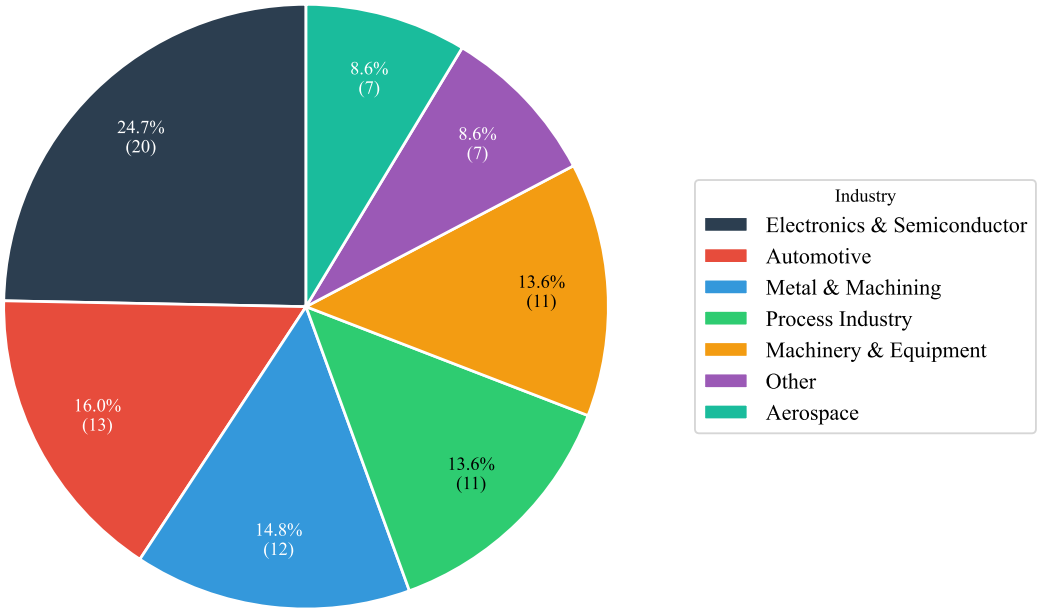



Figure 7: Distribution of industries mentioned in papers, excluding the 124 (60.5%) where the industry was not specified. The Other slice represents one off industries e.g. precast concrete, ship components, nuclear, etc.

Figure 7 shows the distribution of papers across industrial domains. The distribution of papers across industrial domains can also be explored interactively at `https://hmlv-survey.vercel.app/dashboard`.

124 papers (60.5%) did not specify a particular industry. Among those that did, electronics and semiconductor manufacturing is the most represented, followed by metal and machining, automotive, process industry, machinery and equipment, and aerospace. The other slice represents one-off industries (e.g. precast concrete, ship components, nuclear, etc). The high proportion of papers without an industry specification is notable, particularly among the scheduling-focused papers, which often formulate their problems as abstract job-shop instances without grounding them in a specific industrial context.

### 4.4 Technologies and Methods

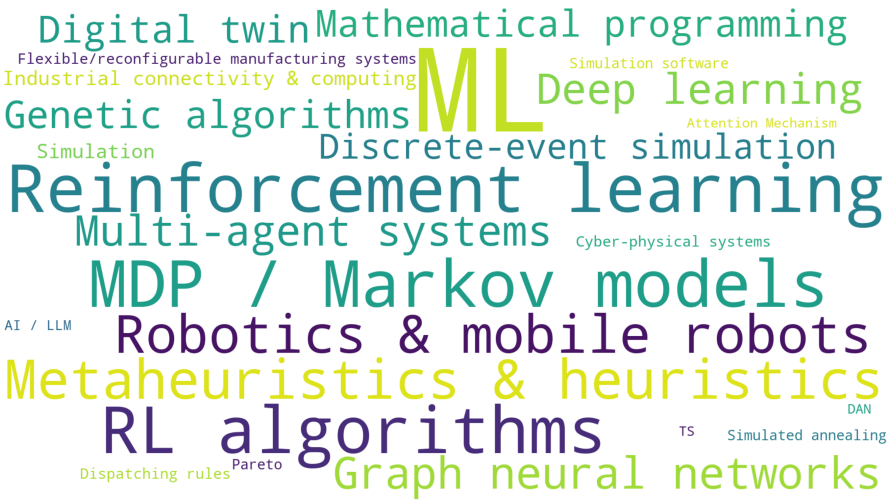



Figure 8: Word cloud of terminology used in the reviewed papers. To generate this figure some method- and technology entries has been grouped, and the relative scale is set to 0.5.

Figure 8 shows the distribution of technologies and methods mentioned across the reviewed papers. The distribution of technologies and methods can also be explored interactively at `https://hmlv-survey.vercel.app/dashboard`.

RL-based approaches are the most frequently mentioned, reflecting their natural fit with the scheduling problems that account for a large share of the corpus. Proximal Policy Optimization (PPO) and Deep Q-Network (DQN) are the most common specific algorithms, while GNNs and Markov Decision Process (MDP) formulations are widely used alongside them for state representation and problem modelling. Priority heuristics, Genetic Algorithms (GAs), and other metaheuristic approaches remain relevant both as standalone methods and as baselines for comparison with learning-based approaches.

Broader terms such as Deep Learning (DL), Machine Learning (ML), and AI overlap considerably with RL, and are often grouped in the literature under the term DRL. On the enabling infrastructure side, CPS, DT, and IoT are the most common, frequently appearing together within Industry 4.0 architectures. Robotics-related technologies, including Automated Guided Vehicles (AGVs), Autonomous Mobile Robots (AMRs), and robot arms, often appear in conjunction with RL, though not necessarily the other way around.

## 4.5 Geographical Distribution

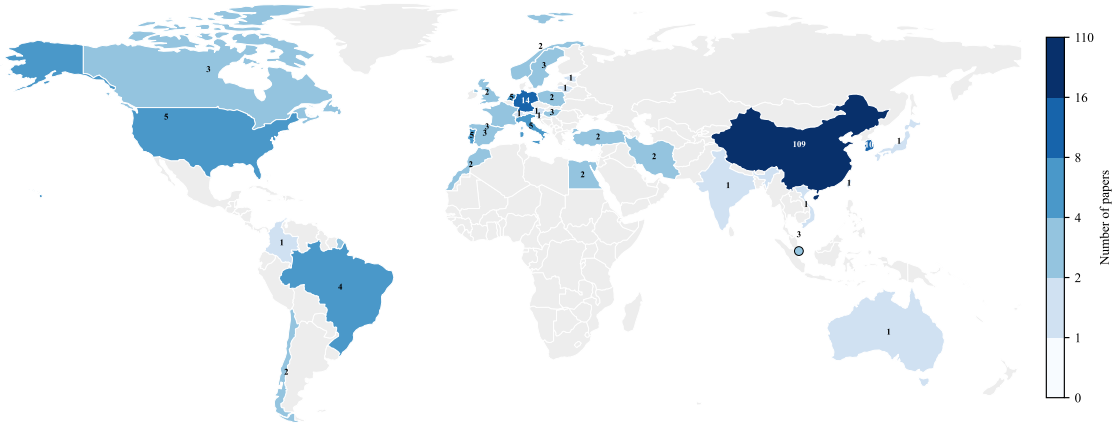



Figure 9: Country distribution of reviewed papers. Based on the country of the research institution of the 1st or primary authors.

Figure 9 shows the geographical distribution of the reviewed papers. China contributes 111 papers, accounting for over half of the reviewed corpus, followed by Germany with 14 papers. Europe collectively forms the second-largest contributing region, with approximately 40 papers across Germany, the Netherlands, Italy, Portugal, Spain, Sweden, and others. Contributions from the Americas remain modest, while contributions from Africa, the Middle East, and Oceania are sparse.

## 4.6 Evaluation Settings

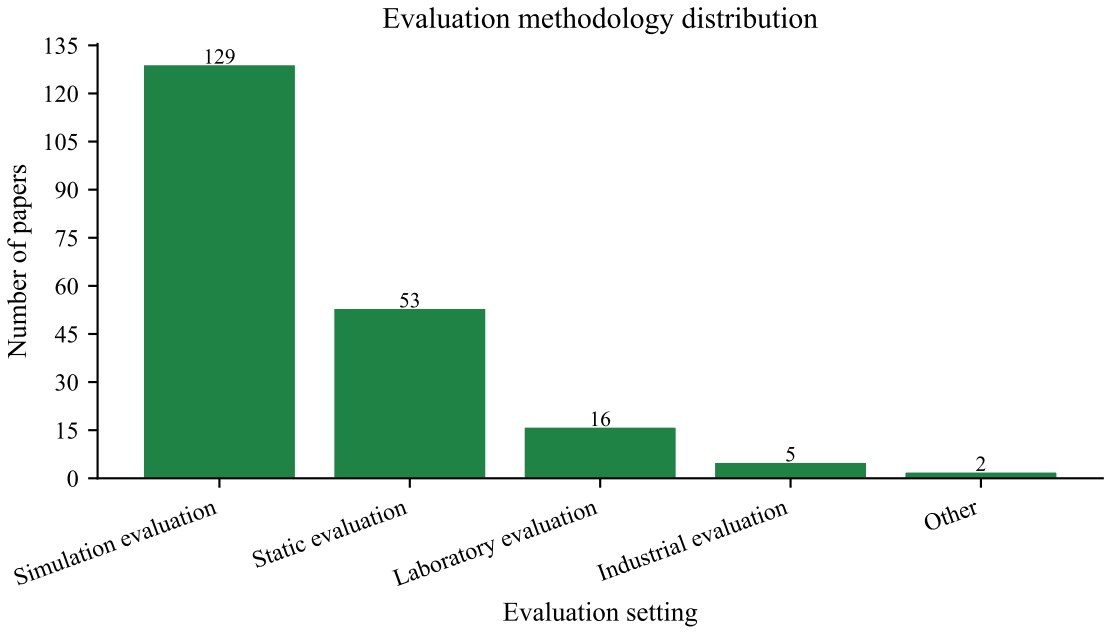



Figure 10: Evaluation setting of the papers.

Figure 10 shows the distribution of evaluation settings across the reviewed papers. The distribution of evaluation settings can also be explored interactively at `https://hmlv-survey.vercel.app/dashboard`.

Simulation-based and static evaluation together account for the vast majority of papers. Few papers evaluate in settings that closely resemble real manufacturing conditions: 16 papers evaluate in a laboratory setting and only 5 papers evaluate in an industrial setting. The high proportion of simulation-based evaluation is largely driven by RL approaches, where a simulated environment is needed for MDP interaction during training. The term *case study* appears frequently in the literature, but actual evaluation within the industrial setting of the case is rare.

### 4.7 Data Sources

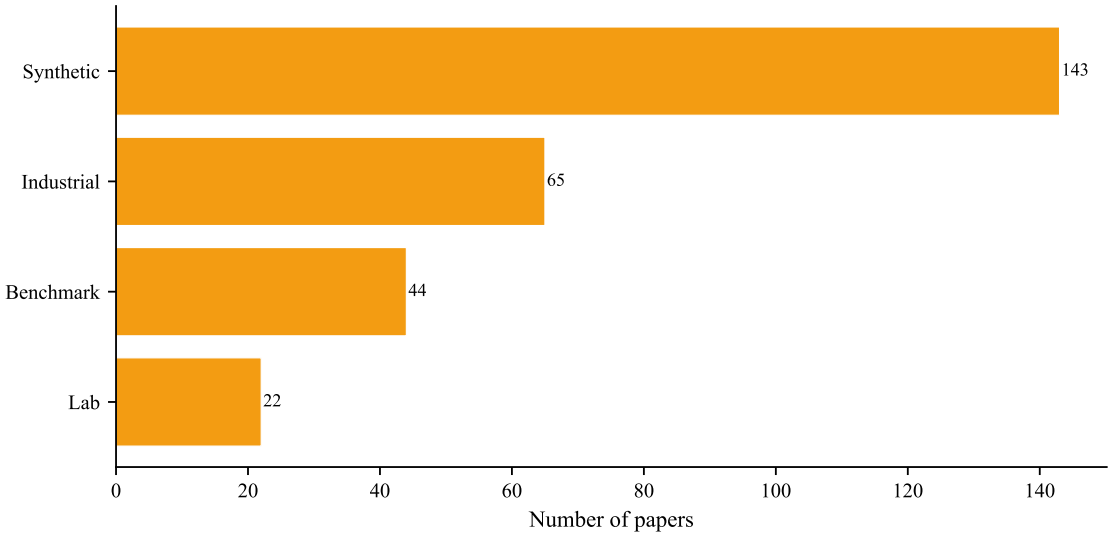



Figure 11: Types of data sources found in the literature.

Figure 11 shows the distribution of data sources used in the reviewed papers. The distribution of data sources can be explored interactively at `https://hmlv-survey.vercel.app/dashboard`

Synthetically generated data is the most common source. Industrial data, including manufacturing layouts, machine parameters, energy usage, operational data, industrial case study data, and process monitoring data, is used in a substantial number of papers. Public benchmarks from various job-shop configurations are also widely used and, in some cases, serve as the foundation for generating synthetic data. Note that papers may use multiple data sources, so these categories are not mutually exclusive. While industrial data is used relatively often, evaluating the approach in an actual industrial setting remains rare.

## 5 Qualitative Analysis

This section presents the qualitative analysis of the reviewed papers, structured by manufacturing level, starting at the process level. The discussion follows the research questions, covering drivers

(RQ1), objectives and effect measured (RQ2), technologies and methods (RQ3), and evaluation settings and limitations (RQ4).

## 5.1 Process level

### 5.1.1 Introduction of Decision Support in HMLV at the Process Level

At the process level, decision support in HMLV manufacturing is primarily focused around the need to execute and adapt individual process steps safely and efficiently despite having a high product variety, frequent changeovers, and uncertainty in both the physical process and its information context. Across the identified process papers, decision support is typically about situated, high-frequency decisions, such as: selecting or parameterising an operation (e.g. a thermal profile), generating a safe robot motion policy during collaboration, provisioning the right information to operators or automation at the right time, and integrating sensing and quality feedback to avoid defects.

The process level covers decision support aimed at individual machines, workstations, or direct human-machine interactions. Of the 205 papers in the corpus, only 14 (6.8%) fall into this category, making it the second smallest manufacturing level reviewed. Despite the low number, four themes emerge: (1) energy-aware single-machine scheduling under time-of-use electricity tariffs [28–34], (2) HRC and robotic task planning [35–38], (3) process quality control and DT based process optimisation, and (4) information provisioning [39–41].

The following subsections discuss the findings for each research question at this level.

### 5.1.2 Drivers at the Process Level

At the process level, the development of decision support is driven by a set of challenges that are specific to the point where products are physically transformed. Five drivers emerge from the reviewed papers.

The most notable finding is that half of the L0 papers focus on energy-aware scheduling, suggesting that energy is seen as one of the few controllable cost factors at the machine level. The remaining drivers, while each represented by fewer papers, highlight that process level decision support in HMLV is not only about optimisation but also about flexibility, information access, and quality assurance.

The majority of papers address single-machine scheduling under time-of-use electricity tariffs [28–34]. These are motivated by the push towards sustainable and greener manufacturing, where scheduling must account for varying electricity prices throughout the day. In HMLV production, variable job durations and frequent changeovers create flexibility in when energy is consumed, making it possible to shift work to cheaper periods. The core challenge is to balance delivery performance against energy cost and, in some cases, carbon footprint constraints.

The second driver is dynamic and unstructured execution conditions in HRC. As Collaborative Robots (cobots) become more common in low-volume production, where full automation is rarely cost-effective, decision support is needed to generate safe motion policies that adapt to the presence and actions of human operators [35].

Third, the high engineering effort per product variant and the need for rapid reconfiguration drives research into task modeling and knowledge-based frameworks for robotic systems. In HMLV

settings, where production tasks change frequently, manually reprogramming robots for each variant is not scalable, creating demand for more flexible approaches [36, 37].

Fourth, incomplete or fragmented information near the process motivates work on information provisioning. Operators and automation systems in configure-to-order environments often lack timely access to the right data, particularly when digital tool chains are not well integrated at the workstation level [41].

Finally, the need for tighter sensor-to-decision loops drives research on zero-defect manufacturing, where real-time quality feedback must be integrated into process level decisions to prevent defects rather than detect them retrospectively [40].

### 5.1.3 Objectives and Measurements at the Process Level

The process level papers pursue three distinct objectives. The largest group targets energy-efficient single-machine scheduling, seeking an optimal balance between delivery performance (tardiness or makespan) and energy cost under time-of-use tariffs [28–34]. Effects are measured quantitatively through energy efficiency and, in some cases, tardiness. Generally, the energy-aware scheduling papers use well-defined quantitative metrics, making comparison across papers straightforward. The remaining papers rely on qualitative demonstrations and case studies, reflecting that process level decision support outside of scheduling is still developing its evaluation standards.

A second objective is safe and adaptive HRC [35], where collision-free robot motion is demonstrated through safety distance and task completion metrics.

Third, several papers aim at reducing the burden of variant complexity, whether through accessible robot programming [36], flexible robotic reconfiguration [37], improved information provisioning [41], or shorter development cycles via DT-based process exploration [39]. Effects here are measured through a qualitative case study assessing usability and robustness.

### 5.1.4 Technologies and Methods at the Process Level

A pattern at the process level is the tight coupling between cyber representations of the process and fast decision logic. The technologies and methods vary, but share common structures.

For HRC, the approaches range from low-level to high-level decision support. Mixed reality combined with DRL is used for execution-time motion policy generation, where the robot must adapt its behaviour in real time based on the operator's position and actions [35]. At a higher level of abstraction, knowledge-driven and model-based AI enables robots to reason about tasks and reconfigure themselves for new product variants without manual reprogramming [37], while no-code task modelling languages aim to reduce the engineering effort needed to specify collaborative workflows for each new variant [36].

DTs appears in two distinct roles at the process level. The first is as surrogate models for process parameter optimisation, where a virtual replica of the physical process (e.g. reflow soldering) is used to explore parameter configurations without disrupting production [39]. The second is as an integration layer for information provisioning, where the DT aggregates data from multiple sources and presents it to operators or automation in a structured way [41]. Closely related, simulation environments combined with RL are used to train operational decision policies, within one process, offline before deploying them [38].

The energy-aware scheduling papers consistently formulate their problems as mixed-integer linear or nonlinear programming models, capturing the trade-off between scheduling objectives (e.g. tardiness, makespan) and energy cost under time-of-use tariffs [28–33]. Because these problems are NP-hard[5], exact solvers are only practical for small instances, and most papers rely on metaheuristic methods such as GAs, artificial bee colony, Simulated Annealing (SA), and greedy construction heuristics to find good solutions within a reasonable time.

### 5.1.5 Evaluation Practices and Limitations of Process Level Decision Support Effects

The reviewed studies at the process level evaluate decision support approaches using several experimental settings, ranging from static optimisation problems to industrial deployment. However, most rely on static, laboratory, and simulation environment settings, while real-world validation remains limited.

The most common evaluation setting used to evaluate decision support in HMLV at the process level is static evaluation [28–34], where the evaluation is performed on a static formulated optimisation problem to produce an optimized schedule. In this setting, the scheduling problem is defined using predetermined job parameters, hourly energy prices over a fixed time horizon, and/or resource constraints. The performance of the proposed approach is evaluated by solving an optimisation instance.

Beyond static optimisation problems at the process level, several studies evaluate decision support approaches within simulation-based or laboratory manufacturing environments. [35] presents an evaluation approach which utilises a simulation environment for training and evaluation, as well as combining this with sim-to-real transfer validation in a laboratory, to illustrate the learned policy generalizability for HRC on several industrial robot arms. Simulation of integrated printed circuit boards and reflow oven models is used in the DT approach proposed by [39] to optimise reflow soldering temperature profiles and improve solder joint quality. Evaluation is performed by comparing the simulated temperature profiles with measurements from the physical reflow process. In contrast, [41] presents a real-world evaluation of a DT through a qualitative, requirement-based assessment in an industrial case study.

Limitations of process level decision support approaches can be classified into four main categories: environmental simplifications, decision support limitations, evaluation and validation constraints, and system-level and deployment limitations, all of which can hinder their use in HMLV manufacturing environments.

Environmental simplification is common in the process level papers. For example, the focus on single-machine operations, which is rarely the case in HMLV productions [28] limiting the approaches' applicability and adaptation. Secondly, a common assumption in the reviewed literature is the use of time-of-use pricing, which relies on predefined energy price intervals. However, modern power systems increasingly employ more dynamic pricing mechanisms, such as real-time pricing, critical peak pricing, and demand response schemes, which are not considered in these approaches [42]. In critical peak pricing and demand response programs, users are incentivised to shift energy

[5]NP-hard (nondeterministic polynomial time) problems are computational problems that are at least as difficult as the hardest problems in NP, meaning that a polynomial-time solution to any NP-hard problem would imply polynomial-time solutions to all problems in NP. They are not necessarily themselves in NP, so a proposed solution may not always be efficiently verifiable. In practice, NP-hardness is used to indicate that a problem is unlikely to have an efficient exact algorithm for all inputs.

consumption outside peak periods. Consequently, the effectiveness of the proposed algorithms may not transfer to such environments.

Furthermore, the production system is often simplified. This is mostly happening through the mathematical model representation, where a deterministic behaviour is assumed, such as fixed processing times, machine availability, and energy prices. In [34], processing time is modelled stochastically, but limited to a single parameter, while other uncertainties such as machine breakdown, machine degradation, or preventive maintenance are not considered [33].

From an evaluation and validation standpoint, a key limitation is that the proposed algorithms are primarily assessed using randomly generated data instances, which might influence the transferability of the results to real HMLV industrial environments [29, 31]. Decision support is largely limited to energy cost minimisation goals, while operational and environment objectives are simplified or excluded [33, 34], despite papers positioning themselves in green and sustainable manufacturing. In addition, broader scheduling metrics such as tardiness, makespan and flow time are not always considered in the energy-aware scheduling approaches.

### 5.1.6 Summary of Decision Support in HMLV at the Process Level

The process level is a less explored area in the corpus, with only 14 papers (6.8%). Research is dominated by energy-aware single-machine scheduling, which accounts for half the papers and is driven by the push towards green manufacturing. The remaining papers address HRC, variant complexity, and defect prevention, each represented by only a few studies.

Across the four research questions, several patterns stand out. The drivers at this level are closely tied to the physical realities of the shop floor: variable energy prices, unsafe or unstructured work conditions, frequent changeovers, and fragmented information. The objectives split into quantitative (energy and tardiness optimisation) and qualitative (safety, usability, information quality), with a corresponding split in how effects are measured. The methods share a common structure of pairing a process model with a decision mechanism, whether that is a mathematical formulation solved by metaheuristics or a DT driving a RL policy. Evaluation remains the weakest point: most papers rely on static or simulated settings with synthetic data, and only one presents an industrial case study, with industrial evaluation.

## 5.2 Line or Cell Level

### 5.2.1 Introduction of Decision Support in HMLV at the Line and Cell Level

At L1, decision support addresses the coordination of interconnected machines, robots, and human operators within a defined manufacturing cell or production line. The cell level deals with the interactions between resources and assets. For example, a robotic arm coordinating with Computer Numerical Control (CNC) machines, or human operators collaborating with cobots in an assembly cell. In L1 the scope remains local: decisions concern a single cell or line rather than plant-wide coordination.

Of the 205 papers in the corpus, 13 (6.3%) fall into this category, making it the smallest manufacturing level. The papers cluster into three themes: scheduling of robotic cells and cluster tools [4, 43–46], HRC and assembly line balancing [47–50], and resource allocation, including both short-term operational decisions and design-level buffer and resource allocation [14, 51]. The

remaining papers address data-driven process parameter recommendation in a HMLV manufacturing environment [52] and manufacturing line scheduling to improve production efficiency and machine utilisation [53]. In the following, we will discuss the findings for each research question at the line or cell level.

### 5.2.2 Drivers at the Line and Cell Level

The drivers at L1 shift from the single-machine concerns of L0 towards coordination problems that arise when multiple resources must work together on varied product configurations.

The most represented driver is the need for efficient scheduling of robotic cells, where multiple robots or machines must be sequenced and timed to minimise idle time, minimise cost and maximise throughput [4, 43–46]. In HMLV production, this is complicated by varying processing times across the product variants and the need to handle frequent changeovers within the cell. A typical industry where these challenges are prevalent is semiconductor manufacturing, where cluster tools are used as part of wafer production. These systems often process heterogeneous wafer types, requiring continuous reconfiguration and varying process flows between batches. Recent work addresses this complexity through optimisation- and learning-based approaches, including mixed-integer programming models and DRL methods [4, 43, 44].

A second driver is the challenge of allocating tasks between human operators and robots in collaborative assembly lines, while also maximising throughput, minimising cycle time and operational costs [47–50]. As product variety increases, the division of labour between humans and robots must adapt to different skill requirements, safety constraints, and variant-specific assembly sequences. This is particularly relevant in HMLV settings, where fixed task assignments are impractical, and many production tasks remain difficult to automate. Unique in this context is the diversity of worker skills, which influences task feasibility and completion time. In [47], this is addressed by incorporating worker skill levels into the task allocation process, modelling workers as proficient, basic, or unable to perform specific tasks.

A third driver is resource allocation under uncertainty, where limited cell resources such as operators, machines, and buffers must be assigned to incoming jobs in response to changing production plans [14, 51]. This is driven by the variability inherent in HMLV production, where demand patterns and processing requirements change frequently.

To summarize, the common driver component across L1 is coordination between: machines in a cell, between humans and robots on a line, or between resources and incoming jobs. While L0 was dominated by single-machine energy optimisation, L1 shifts the focus to how multiple resources (robots, machines, and humans) can work together efficiently despite the variability introduced by high product mix.

### 5.2.3 Objectives and Measurements at the Line and Cell Level

Across L1, objectives are primarily throughput-oriented, with makespan, cycle time, and resource utilisation as the dominant performance metrics. While the HRC papers acknowledge the importance of operator workload, skill and safety, quantitative treatment of human factors such as fatigue, ergonomics, skill development and skill decay remains limited.

At L1, three distinct objectives are pursued. First and foremost, to increase operational efficiency

in robotic line or cell scheduling. This is typically achieved by optimising performance metrics such as makespan and cycle time, which reduces the total time for jobs to pass through the system and thereby increases throughput [4, 43, 44].

Another cluster of papers deals with HRC, where the objectives shift towards workload balancing and task allocation efficiency [47–49]. Common metrics among the HRC papers are cycle time and productivity, often complemented by resource-related measures such as the number of operators or robot cost. For instance, [49] minimises cycle time while reducing the number of human and robot operators, whereas [48] evaluates performance in terms of productivity (completed jobs over time) in a dynamic scheduling context. In addition, some approaches incorporate human-centric and safety-related metrics, such as worker workload (e.g., fatigue) and the proportion of collision-free task execution [50]. The goal is to distribute tasks across human and robotic resources in a way that reduces bottlenecks and, where considered, respecting safety and skill constraints.

The last cluster of papers targets resource allocation and production system buffer design, specifically investigating improved utilisation of operators and machines through simulation-based DSS in the short-term. Resource allocation refers to the process of distributing resources such as machines, operators, and equipment systematically among various processing units or work stations to achieve one or more objectives. This typically has a significant impact on throughput. For instance, [14] demonstrate a hybrid approach combining metaheuristics and ML to minimise bottlenecks and maximise throughput with a particular focus on mobile resources (i.e., operators and AGVs). The study evaluates the approach using the sets of non-dominant solutions and hypervolume metrics describing solution quality, enabling more informed decision-making by identifying optimal solutions and the discovery of resource allocation strategies.

Similarly, [51] proposes a predictive model combined with multi-objective optimisation and lifelong meta-learning to minimise the total number of buffers, operators, and AMRs while maximising throughout, specialising in buffer allocation. In the buffer allocation problem, the intention is to determine the required storage space between the processing units.

### 5.2.4 Technologies and Methods at the Line and Cell Level

RL is used in several robotic cell scheduling papers, with an agent that learns to sequence operations based on real-time cell state feedback [4, 44]. This is well suited to L1 problems because cell states change dynamically as jobs progress through multiple machines, making static schedules less effective than adaptive policies.

Mathematical programming and metaheuristic methods also appear frequently. Mixed-Integer Programming (MIP) is used for cluster tool scheduling with precedence constraints [43], while evolutionary algorithms and metaheuristics are applied to assembly line balancing [47, 49]. Hyper-heuristics are used for integrated process planning and scheduling in reconfigurable manufacturing cells [46]. Monte Carlo Tree Search (MCTS) is used for online decision-making in HRC scenarios where uncertainty in processing times must be handled in real time [48].

Simulation plays a central role at L1, both as an evaluation tool and as a component of the DSS itself. DES is used to model cell behaviour and evaluate scheduling alternatives, and in some cases is combined with data-driven methods to form hybrid DSS [14, 51]. Data-driven approaches also appear in the form of recommendation systems for process parameter selection [52].

### 5.2.5 Evaluation and Limitations at the Line and Cell Level

Evaluation at line and cell level can be grouped into three primary settings: static-based evaluation, simulation-based evaluation, and laboratory-based evaluation.

The largest group of papers falls into static evaluation focusing on optimising production, where decision support approaches are assessed on predefined benchmarks, synthetically generated data or industrial case data [43–45, 47, 49, 52]. In this group, the decision problems are defined in advance, either as optimisation models or data-driven representations of the production system. Typically, the static evaluation papers at this level compare the solution against exact solvers or known heuristic and metaheuristic methods.

The second most common evaluation setting is approaches using simulated environments [4, 14, 46, 51]. These approaches often employ DES or custom environments to represent system behaviour. In some cases, dynamic interactions such as stochastic processing times, resource constraints, or disturbances. Which, compared to the static evaluation setting, gives more realistic conditions and assessment.

Only one paper presents an evaluation in a physical environment, i.e. a laboratory setting [48]. This work demonstrates decision-making during execution through an online decision-making approach implemented in a HRC test cell resembling an industrial setting.

The limitations at the line or cell level share some characteristics with the process level, but also introduce new concerns related to coordination complexity. Several scheduling papers evaluate only on synthetic data with fixed cell configurations, limiting the transferability of results to real HMLV environments where cell and line layouts and product mixes change frequently. The RL-based approaches face scalability concerns: training times increase as the number of machines and jobs in the cell and line grows, and most papers evaluate on relatively small problem instances.

For the HRC papers, a key limitation is the gap between controlled laboratory demonstrations and real production conditions. Safety certification, integration with existing Manufacturing Execution System (MES), and handling of edge cases in human behaviour are rarely addressed.

### 5.2.6 Summary of Decision Support in HMLV at the Line or Cell Level

The line or cell level is the least explored area in the corpus, with 13 papers (6.3%). Research centres around robotic cell scheduling, HRC, and resource allocation, all problems where the core challenge is coordinating multiple resources under the variability of HMLV production.

Compared to the process level, the focus shifts from single-machine optimisation towards adaptive coordination. Methods like RL and MCTS reflect this shift, offering the ability to respond to changing cell and line states. However, objectives remain primarily throughput-oriented, with limited attention to human factors. Evaluation is predominantly simulation-based or static, and the gap between laboratory demonstrations and industrial deployment remains a central limitation. The small number of papers at this level suggests that cell- and line level decision support, including fully implemented DSS, remains an open area with room for future work.

## 5.3 Factory level

### 5.3.1 Introduction of Decision Support in HMLV at the Factory Level

At the factory level, decision support addresses the coordination and scheduling of production across multiple manufacturing lines, cells, or machine groups. This is the most extensively researched level in the corpus, with 151 papers (73.7%), and a strong emphasis on Job-Shop Scheduling (JSP) and their variants. The L2 perspective considers the flow of jobs through multiple processing stages, the assignment of operations to machines across the shop floor, and the sequencing of tasks to optimise overall production performance. Methodologically, the factory level scheduling literature is characterised by two dominant trends: RL-based approaches and metaheuristic optimisation approaches. Classical ML, DL, and broader AI-based methods appear less frequently and are used more often as specialised or emerging alternatives. Table 7 provides an overview of the key characteristics across all four research questions.

### 5.3.2 Drivers at the Factory Level

To address RQ1, each paper is classified according to its primary decision support application area. Scheduling dominates at the factory level, with 118 of 151 papers addressing it as their primary decision area. The driver behind this concentration is the combinatorial complexity of assigning and sequencing jobs across machines in environments where product variety, dynamic job arrivals, and operational uncertainty make static planning impractical.

The most common formulation is the Flexible Job-Shop Scheduling (FJSP), where each operation can be processed on one of several alternative machines (Table 7). This routing flexibility maps directly to HMLV production, where different product variants may require different machine assignments or different configurations of the same machines. Many papers extend the fundamental FJSP with additional constraints such as sequence-dependent setup times, transport between machines, or energy limits [54–56] to better capture real shop-floor conditions. Standard JSP, where each job follows a fixed route, remains the second most common formulation and is often used as a baseline.

A notable trend is the growing focus on dynamic scheduling formulations. Together, Dynamic Flexible Job-Shop Scheduling (DFJSP) and Dynamic Job-Shop Scheduling (DYJSP) account for a substantial minority of the scheduling papers, reflecting the dynamic nature of HMLV production environments. Machine breakdowns, urgent orders, and changing priorities make static schedules difficult to maintain in practice and create a need for approaches that can adapt schedules in real time [57, 58].

The primary driver at the factory level is therefore the combinatorial complexity of scheduling under high product variety. The progression from static formulations towards dynamic variants shows that the field is increasingly driven by the need for approaches that handle uncertainty and variability. This distinguishes the factory level from L0 and L1, where the drivers are tied more closely to physical process conditions and cell level coordination.

Table 7: Overview of factory level (L2) decision support characteristics across the four research questions. Counts indicate the number of papers in each category. Papers may appear in multiple categories. For RQ3, RL and DL are reported as subcategories of ML. The full list of papers per category is available at `https://hmlv-survey.vercel.app/level/L2`.

| RQ | Category | Papers |
|---|---|---|
| RQ1: Application area | Scheduling | 118 |
| | Forecasting & prediction | 7 |
| | Plant layout & reconfiguration | 10 |
| | Visualization & simulation | 9 |
| | Logistics & supply chain | 3 |
| | Quality & maintenance | 4 |
| RQ2: Objectives | Makespan | 88 |
| | Tardiness | 28 |
| | Energy efficiency | 11 |
| | Machine utilisation | 13 |
| RQ3: Enabling tech. | ML (total) | 101 |
| | DL | 10 |
| | RL | 87 |
| | Classical ML | 4 |
| | Metaheuristics | 24 |
| | AI / Large Language Models | 5 |
| | CPS | 7 |
| | IoT | 7 |
| | DT | 14 |
| RQ4: Evaluation | Simulation | 108 |
| | Static evaluation | 28 |
| | Lab case study | 12 |
| | Industrial case study | 2 |
| | Other[a] | 1 |

[a] Proof-of-Concept implementation demonstrating technical feasibility, with no quantitative evaluation.

### 5.3.3 Non-Scheduling Decision Support at the Factory Level

While the factory-level literature has largely focused on scheduling, 33 papers address a diverse set of other decisions. These papers span such a wide range of topics that they do not cluster around a single problem type, and giving each a full treatment across all four research questions would make this section disproportionately long. Instead, this subsection briefly maps the landscape and groups these papers into themes.

The largest theme is production system design and reconfiguration, covering plant layout

optimisation [59–63] and reconfigurable manufacturing systems [64–67]. These papers address how the physical factory is configured or adapted so that jobs can move through it more effectively.

A second theme covers forecasting and prediction, where decision support anticipates future states of the factory. This includes demand forecasting [68–70], lead time prediction [71, 72], bottleneck prediction [73], and power consumption prediction [74]. These approaches can provide inputs to planning and operational decision-making at the factory level.

A third theme is simulation and digital twin support, where the decision support itself is the generation or use of simulation models [75–78]. Related to this, a cluster of papers focuses on monitoring and data infrastructure, addressing how factory data is collected, integrated, and visualised to support human decision-makers [79–82].

The remaining papers cover predictive maintenance [83–85], logistics [86–88], production optimisation [89, 90], material flow control [91], fault diagnosis [92], and product family modularisation [93]. Taken together, these themes show that the factory level supports a broader set of operational decisions. The remainder of section 5.3 focuses on the scheduling literature, where the concentration of research allows for a more structured analysis across the four research questions.

### 5.3.4 Objectives and Measurements at the Factory Level

Makespan minimisation is the default objective at the factory level, appearing in 58.3% of the 151 L2 studies (Table 7). Makespan serves as the standard benchmark metric, particularly for FJSP and JSP papers [94–99].

Tardiness and robustness each appear in a smaller but significant number of papers. Tardiness reflects the importance of on-time delivery in HMLV settings where customers expect reliable due-date compliance [71, 100–102]. Robustness at this level often focuses on generating schedules that still perform well under disruptions such as machine breakdowns or unexpected job arrivals [58, 103, 104]. However, robustness is often claimed qualitatively rather than measured against a formal metric, which makes comparisons between approaches difficult. Energy efficiency and machine utilisation appear less frequently, suggesting that time-based objectives still dominate over resource or sustainability concerns at this level.

A growing trend is multi-objective formulations that combine two or more objectives, most commonly makespan and tardiness minimisation [105–108] or makespan and energy consumption minimisation [55, 109, 110]. These formulations typically use Pareto-based approaches to generate trade-off solutions, giving decision-makers flexibility to choose based on current priorities.

### 5.3.5 Technologies and Methods at the Factory Level

The technology landscape at the factory level is shaped by two main methodological trends. The first and most prominent is RL, which appears in 87 papers and is the dominant approach for factory level scheduling. The second is metaheuristic optimisation, which appears in 24 papers and includes methods such as GAs, SA, and related search-based optimisation techniques. Classical ML, DL, and broader AI-based approaches occur less frequently, indicating that most scheduling research at this level is concentrated around either learning scheduling policies through RL or searching the solution space using metaheuristics.

Within the RL papers, PPO and DQN are the most widely adopted algorithms [57, 59, 84, 111–114]. Their popularity reflects their maturity and availability in standard libraries, which lowers the barrier for applying them to scheduling. It also indicates that these algorithms are well suited to the types of sequential decision-making problems encountered in HMLV manufacturing. GNNs have become an important building block within the RL pipeline, appearing in 28 papers [55, 97, 115–118]. The job-shop structure, consisting of jobs, operations, machines, AGVs, AMRs, precedence relations, and assignment relations, maps naturally onto a graph, making GNNs a good fit for encoding the scheduling state. Multi-agent RL is used in a smaller number of papers to handle scheduling across multiple machines [104, 119, 120].

Metaheuristic approaches form the second major methodological group. These methods treat scheduling primarily as a combinatorial optimisation problem and search for high-quality schedules using algorithmic strategies such as population-based search, neighbourhood exploration, or probabilistic acceptance of worse intermediate solutions. This group includes GAs, SA, and Tabu Search (TS). Compared with RL, metaheuristics do not usually learn a reusable scheduling policy through interaction with an environment. Instead, they iteratively improve candidate schedules for a given problem instance. Some papers also combine both directions, for example, by using metaheuristics to generate initial solutions that are then improved by RL [121, 122].

In contrast, classical ML, DL, and broader AI-based approaches appear less often as primary scheduling methods. When used, they typically support prediction, representation learning, decision assistance, or natural-language interaction rather than forming the main scheduling optimisation mechanism. This suggests that, within the factory level scheduling literature, these methods currently represent more specialised or emerging directions compared with the two dominant paradigms of RL and metaheuristic optimisation.

Separate from the scheduling algorithms themselves, DT, IoT, and CPS appear in a modest number of papers. They typically provide simulation environments for training scheduling agents or real-time data collection from the shop floor [71, 75, 79, 119, 123, 124].

A distinct group of papers integrates AGV or AMR transport time directly into the scheduling problem, reflecting the growing role of automated transport in modern HMLV factories. In some cases, this is further extended to account for operational constraints such as AGV breakdowns and energy consumption [97, 121, 125]. These papers extend classical scheduling formulations by jointly optimising job-machine assignment and AGV routing, either as part of the RL state space [102, 126–129] or through dedicated formulations [56, 97, 125, 130]. The main motivation is that transport capacity becomes a binding constraint when jobs need to move between machines frequently, which is common in HMLV shop floors with routing flexibility. A smaller group uses AGVs beyond scheduling, for process mining [86] or as part of a natural-language scheduling interface [131].

### 5.3.6 Evaluation and Limitations at the Factory Level

Evaluation at the factory level relies mostly on simulation and static settings (Table 7). In simulation-based evaluation, the scheduling agent interacts with a simulated factory that models job arrivals, machine states, and processing times [58, 95, 132–134]. This allows testing under dynamic conditions, but the realism of these simulated environments varies considerably. Static evaluation, where the approach is tested on predefined problem instances with fixed parameters,

is the second most common setting [56, 94, 99, 110, 130]. In some cases, dynamic conditions are partially modelled probabilistically [135]. Static evaluation enables direct comparison between methods on the same instances but does not fully capture the dynamic events of real production.

Laboratory case studies provide more realistic feedback but remain limited in scale [75, 79, 119, 136–138]. Only 2 papers present a full industrial case study, making real-world validation the most significant gap at this level.

Within the factory level papers, the use of benchmark datasets is common for systematic comparison between methods. The identified studies use a range of established public benchmarks, each associated with specific problem types. An overview is provided in Table 8.

Table 8: Public benchmark datasets used in factory level papers. Adapted and extended benchmarks are excluded and discussed separately. A complete overview of the public benchmarks and the corresponding papers using them is available in the appendix A.2.

| **Benchmark dataset** | **Typical problem context** | **Year** | **Original benchmark reference** | **Notes** |
|---|---|---|---|---|
| Brandimarte (mk) | FJSP | 1993 | [139] | Classic flexible job shop benchmark. |
| Fisher-Thompson (ft) | JSSP | 1963 | [140] | Standard JSSP benchmark. |
| Lawrence (la) | JSSP | 1984 | [141] | Appears also in derived variants (e.g., via Hurink). |
| Applegate-Cook (orb) | JSSP | 1991 | [142] | Standard JSSP benchmark. |
| Adams-Balas-Zawack (abz) | JSSP | 1988 | [143] | Standard JSSP benchmark. |
| Taillard (ta) | JSSP | 1993 | [144] | Widely used benchmark set. |
| Demirkol-Metha-Uzsoy (dmu) | JSSP | 1998 | [145] | Extended JSSP benchmark instances. |
| Storer-Wu-Vaccari (swv) | JSSP | 1992 | [146] | Standard JSSP benchmark. |
| Yamada–Nakano (yn) | JSSP | 1992 | [147] | Less frequently used benchmark. |
| Hurink (edata/rdata/vdata/sdata) | FJSP | 1994 | [148] | Derived flexible job shop benchmarks. |
| Dauzère | FJSP | 1997 | [149] | Flexible job shop benchmark. |
| Kacem | FJSP | 2002 | [150] | Widely used FJSP benchmark. |
| Fattahi / Homayouni-Fontes | FJSP | 2007 | [151] | Flexible job shop benchmark. |
| Behnke | FJSP | 2014 | [152] | Less standardized benchmark. |
| Deroussi-Norre | FJSP | 2010 | [153] | Specialized benchmark (AGV context). |
| Boccia | AGV scheduling problem | 2023 | [154] | Limited context information. |

Several studies also develop or extend existing benchmark instances by adding distributed, dynamic, or domain-specific constraints. These are excluded from Table 8, but their prevalence highlights the lack of standardised benchmarks for more complex problem settings. This may partly be because many established benchmarks are relatively old and are not specifically tailored to HMLV manufacturing. Only one recent benchmark from 2023 was identified in the corpus. However, more recent domain-specific benchmarks exist outside the corpus, such as SMT2020 [155], which includes a dedicated low-volume/high-mix (LV/HM) wafer fabrication model with features representative of modern semiconductor manufacturing environments. This indicates that, while

more recent domain-specific benchmarks exist, they are not widely adopted in the broader HMLV scheduling literature.

The main limitations regarding evaluation can be grouped into four areas. First, reproducibility is limited because many papers do not provide source code or sufficient implementation details for replication. Second, method comparison is often narrow, with papers comparing against basic dispatching rules or earlier versions of the same method rather than current alternatives. Third, scalability remains uncertain because most approaches are tested on small to medium instances, while performance on the larger problems found in real factories is unclear. Fourth, integration with enabling infrastructure, which is necessary for fully implemented DSS, is rarely demonstrated. Despite frequent mention of Industry 4.0 in the broader literature, actual integration of RL-based or metaheuristic scheduling approaches with DT, IoT, or MES systems is limited. Most scheduling papers train and evaluate in custom simulation environments or predefined benchmark settings rather than on data from DTs or shop-floor systems, and how such approaches would connect to existing MES or shop-floor control systems is rarely discussed.

### 5.3.7 Summary of Decision Support in HMLV at the Factory Level

L2 is by far the most researched area in the corpus. Scheduling dominates, with FJSP as the largest single problem type and a clear trend towards dynamic formulations. Makespan is the default objective, although multi-objective formulations combining it with tardiness or energy consumption are growing. Methodologically, the literature is concentrated around two main trends: RL, which is the dominant approach, and metaheuristic optimisation, which forms the second major group. Classical ML, DL, and broader AI-based approaches are less frequent and appear mainly as specialised or emerging alternatives. Evaluation remains the weakest aspect, with only 2 industrial case studies across 151 papers. The main gaps are the disconnect between benchmarking and industrial validation, the limited attention to objectives beyond makespan, and the absence of integration with the enabling infrastructure that would be needed for real deployment.

## 5.4 Enterprise and Supply Chain Level

### 5.4.1 Introduction of Decision Support in HMLV at the Enterprise and Supply Chain Level

At L3, decision support addresses decisions that extend beyond the boundaries of the individual factory to what flows in and out of it. This level covers multi-factory and distributed scheduling, internal and external logistics, demand forecasting, supplier and service selection, and strategic alignment between product design, manufacturing systems, and supply chain partners. 27 papers (13.2%) in the corpus address this level. The decisions at this level are typically characterised by uncertainty in demand, supplier reliability, and coordination across distributed sites.

### 5.4.2 Drivers at the Enterprise and Supply Chain Level

Research at this level is driven by two roughly equal concerns: distributed operational coordination and strategic alignment across organisational boundaries. The majority of the papers address Distributed Job-Shop Scheduling (DIJSP) where jobs flow across multiple factories or reconfigurable

workshops [156–166].The driver here is the coordination challenge that arises when jobs need to be allocated across geographically distributed sites with different capabilities, capacities, and constraints.

A second cluster of papers addresses logistics and production-logistics synchronisation [167–172]. These papers are motivated by the need to coordinate production with warehousing, transportation, and distribution. A recurring motivation in this group is resilience against disruptions, a concern that has gained traction in recent years alongside the inclusion of resilience in the EU's Industry 5.0 definition [173]. Yang et al. [167] target lower logistics cost and higher resilience by integrating logistics and manufacturing decision making, while Zhang et al. [170] propose a blockchain and DT-based framework for trusted synchronised decision-making across a social production logistics system.

The remaining papers address a diverse set of strategic and supporting decisions: demand forecasting for MTO environments [174, 175], supplier performance prediction in ETO supply chains [176], customisation level classification [177], selection of Manufacturing as a Service (MaaS) partners [178], plant layout design [179], supply chain evaluation [180], coordination between product design and production [181], and internal supply chain planning [182]. These topics do not cluster around a single problem type, reflecting the breadth of decisions that arise at this level.

The primary drivers are distributed scheduling across multiple sites and production-logistics coordination, with resilience emerging as a motivating theme for the logistics group. Together, scheduling and logistics account for roughly two thirds of the literature at this level, with the remainder covering strategic and supporting decisions.

### 5.4.3 Objectives and Measurements at the Enterprise and Supply Chain Level

Makespan minimisation is a common objective at this level, particularly in the distributed scheduling papers, where it is used either as the sole objective or combined with energy, tardiness, or cost minimisation [156–158, 160, 161, 164, 166]. Tardiness and due-date compliance also appear as objectives, typically in scheduling and logistics papers [159, 168, 182].

A distinctive feature at this level is the prominence of cost-based objectives. Total system cost, combining production, storage, and distribution costs, is used in several of the logistics and supply chain papers [162, 169, 170, 181], reflecting the multi-stage nature of the decisions at this level. Multi-objective formulations are common, particularly in logistics and scheduling papers that balance cost against time, energy, or maintenance criteria.

Total system cost spanning production and distribution is a defining objective at this level, reflecting the multi-stage nature of the decisions. Makespan and tardiness remain relevant, and multi-objective formulations that balance several of these concerns at once are common.

### 5.4.4 Technologies and Methods at the Enterprise and Supply Chain Level

ML is the most common technology at this level, appearing in several papers [69, 168, 174–178], where it is applied to supplier selection, customisation classification, forecasting, and MaaS selection. RL also appears in multiple works [156–161], mostly in distributed scheduling, and in some cases, in multi-agent settings [164, 166]. DT technology typically appears in the logistics cluster as the underlying infrastructure for coordination and real-time control [162, 169–172, 182], often in

combination with other technologies.

On the methods side, metaheuristic optimisation approaches are prominent, with GA, SA, TS, and adaptive large neighbourhood search [162, 169, 179, 181]. DQN and other RL algorithms are used mainly in the scheduling papers [156, 157, 159, 160], which in some cases is combined with metaheuristic algorithms [158, 161]. Fuzzy methods and Multi-Criteria Decision-Making (MCDM) techniques appear in papers addressing strategic or selection decisions [167, 178], consistent with the structural nature of those problems.

MaaS warrants a short note of its own. It has gained attention as a way to improve resilience through strong collaboration and communication across the value chain. Research found here focuses on tools for selecting appropriate MaaS partners, such as the contracting and evaluation methodology proposed by Gong et al. [178] based on information content measurement and decision tree learning.

The technology landscape is mixed, with ML and DTs playing distinct roles. ML is used in distributed scheduling, forecasting, and selection decisions, while DTs is primarily applied in logistics coordination. Methods are correspondingly varied, with metaheuristic approaches playing a significant role alongside RL-based and other learning-based methods.

### 5.4.5 Evaluation and Limitations at the Enterprise and Supply Chain Level

Evaluation relies predominantly on static evaluation and simulation, with only two industrial deployments [171, 180]. Static evaluation is common in the scheduling and metaheuristic papers [161, 163, 169, 179], while simulation is used across scheduling, logistics, and forecasting contexts [156, 157, 162, 170, 175, 182]. Data sources are a mix of industrial data [167, 168], synthetic data [160, 163], and combinations of both [169, 179].

The limitations reported here fall into four areas. Reproducibility is limited, as most papers do not provide source code. Method comparison is often restricted to a small set of baselines rather than current alternatives. Integration with enterprise information systems such as Enterprise Resource Planning (ERP) or MES is rarely demonstrated, despite being essential for practical deployment.

### 5.4.6 Summary of Decision Support in HMLV at the Enterprise and Supply Chain Level

The enterprise and supply chain level covers a broad range of decisions, from distributed scheduling to logistics coordination, supplier selection, and strategic alignment. Scheduling and logistics together account for most of the literature, while the remaining papers address a long tail of specialised topics. Makespan remains a common objective but is joined by total system cost and tardiness, and multi-objective formulations are common. The technology landscape is mixed, with ML applied in distributed scheduling, often through RL-based approaches, as well as in forecasting, and selection decisions, while DTs are primarily used in logistics coordination.

# 6 Discussion

This work reviews 205 papers on decision support in HMLV manufacturing. The reviewed literature is organised across four manufacturing levels: process (L0), line and cell (L1), factory (L2), and enterprise and supply chain (L3). Across these levels, the papers are analysed through four research questions addressing drivers (RQ1), objectives (RQ2), technologies and methods (RQ3), and evaluation practices (RQ4). This cross-level perspective allows the review to compare how decision support changes across different manufacturing contexts and organisational layers.

From a research perspective, the review maps existing decision support approaches and their main characteristics across manufacturing levels. From a practitioner perspective, it provides an overview of methods and technologies relevant to different decision-making contexts in HMLV manufacturing. An interactive overview of the reviewed papers is available at `https://hmlv-survey.vercel.app/`, where researchers and practitioners can explore the corpus by manufacturing level, decision area, technology, and publication year.

## 6.1 Summary of Key Findings and Research Gaps

The analysis of the reviewed literature reveals several structural imbalances in how decision support for HMLV manufacturing is currently approached. Most notably, the field is heavily dominated by scheduling related research. This is unsurprising, as scheduling represents one of the most acute operational challenges in HMLV environments, where frequent product changeovers, small batch sizes, and variable processing times create highly combinatorial and dynamic decision problems. However, within the scheduling related papers only three of the JSSP papers directly name HMLV as the desired setting [58, 183, 184]. The remaining articles all describe potential solutions which have either been benchmarked against using HMLV scenarios without explicitly mentioning HMLV, or do not describe constraints to the solutions which would imply that the solution(s) do not work in HMLV scenario. Consequently, a large portion of literature focuses on developing increasingly sophisticated optimisation and learning-based approaches for scheduling and rescheduling tasks.

In contrast, other decision support domains receive significantly less attention. Areas such as logistics and supply chain management (6.3%), plant layout and reconfiguration (6.8%), and forecasting and prediction (6.3%) are comparatively under-represented. While these domains often operate on longer time horizons or serve as inputs to scheduling decisions, they nevertheless play a critical role in overall factory performance. Particularly notable is the limited focus on maintenance and quality decision support, which together account for only 2.9% of the reviewed papers, despite their importance in HMLV settings where defects and equipment failures can have disproportionately high consequences. Similarly, human-robot collaboration (2.9%) remains a niche topic, suggesting that human-centric aspects of decision support have not yet been fully integrated into mainstream research.

At the same time, the distribution of research across manufacturing levels shows a strong concentration at the factory level (L2), which accounts for over 73.7% of the total corpus. This reflects the central role of factory-level decision-making in coordinating complex production environments. However, a closer inspection reveals that much of this work is still centred in scheduling problems, indicating that factory-level decision support is often interpreted narrowly as scheduling optimisation. In contrast, lower levels such as L0 and L1 are relatively under-represented, possibly due to the

increased complexity of integrating DSSs with physical equipment and real-time sensor data. Similarly, higher-level decision support at L3, while present, remains a minority, suggesting that HMLV-specific challenges are primarily addressed within the factory boundary rather than at the enterprise and supply chain level.

Furthermore, most evaluations in the corpus are conducted outside actual industrial production environments. Across all levels, studies primarily rely on simulation-based evaluation (62.9%) or static, model-based settings (25.9%). Although case studies are frequently reported, they rarely involve validation within the real production environment from which the case is derived. At the same time, a relatively high proportion of studies utilise industrial data (31.7%), indicating that real-world data is often available even when deployment or validation in operational settings is limited.

Despite this, a notable observation in the high proportion of studies that do not specify a concrete industrial application domain. Many contributions, particularly within scheduling, formulate their problems in abstract terms such as job-shop or flow-shop models without grounding them in a specific industrial context. While this abstraction supports generalisability and methodological development, it raises concerns regarding practical applicability, as real-world HMLV manufacturing environments are often characterised by domain-specific constraints related to materials, regulations, and product complexity. Among the studies that do provide an industrial context, sectors such as electronics and semiconductor manufacturing are prominently represented, reflecting their inherent HMLV characteristics. In contrast, industries such as aerospace, despite being a canonical example of HMLV production, appear comparatively under-represented within the corpus.

From a geographical perspective, the distribution of research is similarly uneven. A significant portion of the literature originates from China, followed by strong representation from European countries such as Germany. The distribution likely reflects the influence of national industrial strategies, including initiatives such as ”Made in China 2025” and ”Industry 4.0”, which have prioritised the development of smart manufacturing technologies and AI-driven decision support. This suggests that research activity in this domain is not only driven by technical challenges but also by broader policy and economic factors.

Finally, a noticeable increase in the number of publications from 45 in 2024 to 81 in 2025 indicates a rapidly growing research interest in decision support for HMLV manufacturing. This growth is likely supported by the maturation of enabling technologies such as ML, DTs, IoT, and CPS, which have lowered the barrier for developing advanced DSSs. However, as the subsequent subsection will demonstrate, this growth has not been uniform across research areas, and several important gaps remain in terms of practical applicability, system integration, and coverage of non-scheduling decision domains. At the enterprise level L3, the relative scarcity of contributions, combined with the limited integration with enterprise information systems, suggests that decision support beyond the factory boundary remains underexplored. In particular, the integration of strategic decision-making with real-time digital infrastructure represents a potential direction for future research.

### 6.1.1 Summary of the Findings at the Manufacturing Levels

At the **process level** (14 papers, 6.8%), decision support is closely tied to process understanding. Every approach pairs a model of the process, whether a DT model, a simulation environment, a

knowledge base, or a mathematical formulation, with a decision mechanism that acts on that model. Research is dominated by energy-aware single-machine scheduling under time-of-use tariffs, with the remaining papers addressing HRC, variant complexity, and defect prevention. Methods are predominantly metaheuristic, and evaluation relies almost entirely on static optimisation instances with synthetic data. Only one paper presents an industrial case study at this level, which points to a clear gap between how approaches are developed and the conditions they would need to operate in.

At the **line and cell level** (13 papers, 6.3%), the focus shifts from single-machine optimisation to multi-resource coordination: scheduling robotic cells, balancing human-robot assembly lines, and allocating resources under uncertainty. Compared to L0, there is a stronger presence of adaptive methods such as RL and MCTS, which fits the more dynamic nature of cell-level coordination, where decisions must respond to changing states rather than solving fixed problem instances. However, objectives remain primarily throughput-oriented, with limited quantitative treatment of human factors such as fatigue, skill development, and ergonomics. This is a notable gap, as HMLV production relies heavily on skilled operators, particularly when handling disturbances in dynamic environments.

At the **factory level** (151 papers, 73.7%) represents the largest fraction of the corpus and is heavily centred on scheduling, with the FJSP as the largest single problem type and a clear trend towards dynamic formulations. Within the broader ML category, RL is the dominant scheduling approach, with PPO and DQN as the most common algorithms and GNNs increasingly used for state representation [55, 57, 112–114]. This is a clear shift from the metaheuristic-dominated lower levels. Makespan minimisation remains the default objective, though multi-objective formulations combining it with tardiness or energy consumption are growing [55, 105–110, 166]. Beyond scheduling, a diverse set of decisions at the factory level are addressed including plant layout, forecasting, logistics, and predictive maintenance, but these do not cluster around a single problem type. On the methodological side, factory-level scheduling is the most mature area in the corpus: the combination of RL with GNNs has produced a well-developed research pipeline with established benchmark datasets and systematic comparison across methods. However, the main gaps at this level are the disconnect between this benchmarking progress and industrial validation, the limited attention to objectives beyond makespan, and the absence of integration with enabling infrastructure such as DTs, IoT, and MESs that would be needed for real deployment.

At the **enterprise and supply chain level** (27 papers, 13.2%), research is driven by distributed scheduling across multiple sites and production-logistics coordination, with resilience emerging as a motivating theme. The technology landscape is mixed: ML is concentrated in distributed scheduling, forecasting and selection decisions, and DTs in logistics coordination. Cost-based objectives are more prominent at this level compared to the lower levels, alongside makespan and tardiness, which fits the multi-stage nature of these decisions. There are relatively few contributions at this level, and integration with enterprise information systems such as ERP and MES is limited. This suggests that decision support beyond the factory boundary remains underexplored, particularly when it comes to connecting strategic decision-making with real-time digital infrastructure.

### 6.1.2 Key Findings and Research Gaps

**Evaluation and the sim-to-real gap.** A limitation across the reviewed literature is the strong reliance on simulation-based evaluation. This is particularly evident in RL approaches for flexible job shop scheduling, where performance improvements are demonstrated under controlled experimental conditions [96, 185–187]. These studies typically evaluate policies using benchmark datasets or synthetic job shop environments, allowing for clear comparisons in terms of makespan, tardiness, or energy consumption. Although such results are valuable for methodological development, they provide only limited insight into how these approaches perform under real manufacturing conditions, where uncertainty, noise, and system variability are unavoidable.

Importantly, this limitation is not limited to RL. Hybrid and optimisation-based approaches exhibit similar evaluation patterns, relying heavily on structured problem instances and static assumptions in a static evaluation setting [107, 188, 189]. Even when dynamic elements are included, such as stochastic job arrivals or machine breakdowns, they are often modelled in simplified ways that do not fully reflect real production complexity. As a result, the literature demonstrates strong progress in solving formalised decision problems, but less progress in validating whether these solutions can be effectively deployed as DSSs in operational environments. This lack of real-world validation is further reflected in the high proportion of studies that do not specify a concrete industrial application domain. Many contributions formulate their problems in abstract terms without grounding them in specific production environments, which while supporting generalisability, limits insight into how these approaches perform under real operational constraints. Across different manufacturing levels, a consistent pattern emerges in how DSSs are developed. Most approaches rely on an explicit model of the system, such as a simulation environment, a mathematical formulation, or a digital representation, paired with a decision mechanism that operates on this model. While this separation enables controlled experimentation, it also reinforces the gap between modelled environments and real-world systems, where process understanding is often incomplete, noisy, or continuously evolving.

The core issue is not merely the absence of real-world experiments, but a broader sim-to-real gap. Many approaches provide complete and accurate system state information, stable objective functions, and seamless execution of decisions. In practice, manufacturing environments, especially in HMLV contexts, are often characterised by incomplete data, frequent disturbances, and the need for human oversight. DSSs must therefore be robust, interpretable, and adaptable, qualities that are difficult to assess through simulation alone. Without addressing these aspects, there is a risk that reported performance gains do not translate to real operational environments. This gap between development and deployment is further reinforced by the evaluation practices observed by the literature. At both L0 and L2, the vast majority of studies rely on static benchmark instances or simulation-based experiments, often under deterministic assumptions. Only very limited number of works evaluate their approaches in real production environments. As a result, while the literature demonstrates strong methodological progress, there is comparatively little evidence of how these approaches perform under real-world conditions characterised by uncertainty, incomplete data, and operational constraints.

That said, a smaller subset of the literature begins to address this gap by moving toward more realistic and integrated evaluation settings. For example, DT-based and cyber-physical

approaches explicitly consider real-time synchronisation and interaction between physical and virtual systems [137, 190, 191]. These works suggest a shift from purely offline optimisation toward more deployment-oriented decision support. However, such approaches remain relatively underrepresented. Future research should therefore prioritise validation strategies that go beyond simulation, including real-time data integration, laboratory-scale implementations, and industrial case studies with in-production evaluation, in order to better assess the practical applicability of proposed methods.

**Evaluation and the sim-to-real gap.** Across all levels, evaluation remains the weakest aspect of the literature. Most studies rely on simulation-based (62.9%) or static (25.9%) settings. Only 5 papers present industrial case studies: one at L0 [41], two at L2 [62, 71], and one at L3 [171, 180]. Although 31.7% of studies use industrial data, validation within real production environments is rare. A high proportion of studies (60.5%) do not specify a concrete industrial application domain, and instead formulate problems in abstract terms. While this supports generalisability, it limits insight into practical applicability. The core issue is a broader sim-to-real gap. Even methodologically strong contributions, such as recent RL-based scheduling approaches [96, 185] and hybrid optimisation methods [188, 189], are evaluated only in simulation or on static benchmarks. RL studies typically evaluate policies using benchmark datasets or synthetic job shop environments, allowing clear comparisons, such as in terms of makespan, tardiness, or energy consumption. Similarly, hybrid and optimisation-based approaches often rely on structured problem instances and static assumptions. Even when dynamic elements are included, such as stochastic job arrivals or machine breakdowns, they are often modelled in simplified ways that do not fully reflect real production complexity. These controlled settings typically provide complete system state information, stable objective functions, and seamless execution of decisions. Real HMLV environments, in contrast, are often characterised by incomplete data, frequent disturbances, and the need for human oversight. This gap is reinforced by a common pattern across all levels: an explicit system model is paired with a decision mechanism that operates on it. While this separation enables controlled experimentation, it creates a gap between the research setting and real-world conditions where process understanding is often incomplete or continuously changing.

**Under-explored human-in-the-loop decision support.** Most decision support approaches and DSSs in the corpus do not involve the human in the decision loop. Human factors are acknowledged but rarely integrated as a contributing agent. Instead, humans are mainly modelled as resources with attributes such as skill level, task completion time, fatigue, or ergonomic constraints within optimisation and scheduling frameworks [35, 47, 50]. This is especially clear at L1, where operators are included in models but mainly as resources to be allocated, not as agents who contribute to decisions. For example, [47] develops an approach that orchestrates tasks between operators and robots, but the decisions are made entirely by the algorithm with no operator feedback. Similarly, [48] presents an approach for real-time scheduling that, while operating in a collaborative setting, does not include human input in the decision-making process. Optimisation objectives are also largely system-centric, and human-related factors such as workload, cognitive effort and decision contribution are rarely modelled explicitly. HRC accounts for only 2.9% of the reviewed papers. Full automation is not always practical in HMLV environments. The high variability, frequent disturbances, and changing requirements make it challenging, and increasing automation often brings higher costs and reduced flexibility [49]. Humans can more easily adapt to

new situations, handle unexpected events, and apply contextual understanding where data is scarce or incomplete [50].

**Digital twin enabled decision support.** While RL dominates the reviewed literature, DT-enabled decision support emerges as a promising but comparatively underexplored direction. Unlike traditional optimisation approaches that operate in static or simulated environments, DTs provide a dynamic and continuously updated representation of the physical manufacturing system. This enables not only improved system visibility, but also the possibility of real-time simulation, prediction, and decision-making. As such, DTs shift the focus from isolated optimisation tasks toward integrated, system-level decision support. The emergence of DT-based approaches can also be linked to the broader growth in decision support research observed in recent years, driven by advancements in enabling technologies such as real-time data integration, simulation platforms, and AI methods. Despite this growth, however, DT applications remain relatively under-represented compared to more established optimisation-based approaches.

Several studies in the reviewed corpus highlight the potential of this approach. At the architectural level, DT integration has been explored as a structured methodology for connecting simulation models with real-time production systems, enabling synchronisation between physical and cyber layers [190]. Similarly, research on DT-based shop-floor systems and autonomous decision-making mechanisms emphasises the role of DTs in enabling closed-loop control and real-time responsiveness [192, 193]. These contributions are important because they frame decision support not just as an optimisation problem, but as part of a broader cyber-physical system.

At the operational level, DTs have been applied directly to scheduling and production control. For example, cognitive DTs have been proposed for multi-objective scheduling, enabling more context-aware and adaptive decision-making [137]. Other works combine DTs with RL to create hybrid systems capable of both learning and real-time adaptation [191]. Additionally, simulation-driven scheduling within DT environments demonstrates how traditional modelling approaches can be extended into continuously updated DSSs [194]. These studies collectively suggest that DTs can serve as a bridge between simulation-based optimisation and real-world applications.

Despite these promising developments, DT approaches remain relatively scarce compared to the extensive body of scheduling-focused research. This indicates that the field has not yet fully explored its potential. Given their ability to support real-time data integration, adaptive decision-making, and system-wide visibility, DTs appear particularly well-suited for HMLV manufacturing environments.

One potential explanation for the limited adoption of DTs among the decision support studies identified can be caused by computational requirements limiting the response time for near- or real-time simulation use. DTs that closely reflect production systems are used to model and simulate complex interactions between entities, increasing computational demands in the systems. As highlighted, high-fidelity modelling, multi-domain integration, and continuous data synchronisation impose computational burdens and challenge real-time performance [195–197] as also observed in prior work. This introduces a fundamental trade-off between model fidelity and computational efficiency, where increasing realism directly impacts scalability and responsiveness [196]. As a result, achieving real-time or near-real-time decision support remains a non-trivial challenge, often requiring distributed computing strategies to mitigate latency and computational bottlenecks. These challenges become particularly evident in applications involving closed-loop control and

what-if scenario analysis, where repeated simulation and rapid evaluation are required under strict time constraints. In such settings, reliance on high-fidelity models can hinder responsiveness and limit practical deployment. This limitation is also reflected in practice, where computationally intensive simulation models are replaced by simpler data-driven approximations due to prohibitive runtime [51].

**Maintenance and quality decision support.** In HMLV environments, where production volumes are low and products are highly customised, a single defective unit or an unexpected equipment failure can represent a large share of total output and cost. This makes reliability and defect prevention operationally critical, but the research attention given to these areas does not reflect their practical importance. Some studies do incorporate maintenance into scheduling frameworks [161, 198–200], but maintenance is typically treated as a secondary objective or constraint rather than as a primary decision-making domain. In practice, maintenance decisions involve trade-offs between immediate production goals and long-term equipment health, depend on incomplete or uncertain information about machine condition, and require coordination across multiple stakeholders. These challenges are not fully captured by embedding maintenance within scheduling optimisation. A similar pattern holds for quality: only a small number of studies explicitly address it, such as frameworks for assembly quality inspection in small-batch production [201] and methods for detecting process variability [202]. No reviewed paper connects process-level anomaly detection with causal tracing across the production chain to support factory-level quality decisions. This suggests that quality-related decision support remains fragmented rather than integrated.

## 6.2 Future Research Directions

Based on the findings and gaps identified above, five directions for future research are proposed.

**Industrial validation and closing the sim-to-real gap**. The most pressing need across all manufacturing levels is validation that goes beyond simulation. Future research should prioritise evaluation strategies that include real-time data integration, laboratory-scale implementations, and industrial case studies with in-production evaluation. However, in-production evaluation can be limited by implementation cost, technical integration, restricted access to operational environments, and the risk of disrupting production. A practical progression is therefore to move from realistic simulation and historical industrial data to laboratory-scale implementations, followed by shadow-mode evaluation, where recommendations are recorded but not executed, and finally bounded in-production trials with appropriate fallback procedures. A smaller part of the literature already moves in this direction through DT-based and cyber-physical approaches that consider real-time synchronisation between physical and virtual systems [137, 190, 191]. Expanding such deployment-oriented approaches and developing standardised protocols for industrial validation would help close the gap between methodological progress and practical applicability. Because evidence of real-world applicability is necessary for the adoption of the other research directions, industrial validation is considered the highest immediate priority.

**Human-in-the-loop decision support.** Future research should explore how human input can be included during execution, for example through interactive or feedback-based DSSs that allow operators to influence decisions in real time. It would also be valuable to move beyond

static representations of humans towards dynamic modelling of learning, experience, and changing performance over time, such as skill improvement and degradation. It may also be relevant to investigate hybrid approaches, where algorithmic methods and human judgement are combined rather than treated separately, particularly for HMLV environments where full automation is impractical and human flexibility is a key asset. Practical challenges include operator trust, cognitive workload, differences in experience, responsibility for decisions, and the safe handling of overrides. Initial studies could therefore evaluate explainable recommendations that require operator approval before execution. In addition to production performance, these studies should measure operator acceptance, override frequency, decision time, and workload. Because HMLV production already depends heavily on human flexibility and contextual knowledge, human-in-the-loop decision support is considered a near-term priority.

**Digital twin enabled and infrastructure-integrated decision support.** Given the potential of DTs for real-time simulation, closed-loop control, and system- level visibility, future research could focus on hybrid DT frameworks that integrate learning-based, optimisation-based, and data-driven methods [63]. Addressing the fidelity-efficiency trade-off through lightweight or surrogate models, edge-computing strategies, and selective twinning approaches would help make DT- enabled decision support more practical. Future research should also explore the role of DTs in validation, cross-level integration, and deployment-oriented decision support. More broadly, closing the gap between algorithmic advances and enabling infrastructure, including DTs, IoT, and MESs, is needed to move from research prototypes to deployable DSSs. Initial applications should focus on bounded decision problems and evaluate whether the operational benefits justify the required integration and model-maintenance costs. This is considered a medium-term enabling direction with potentially high impact where real-time system visibility is required, although not every decision-support application requires a complete high-fidelity DT.

**Maintenance and quality as primary decision domains.** Rather than embedding maintenance and quality as secondary constraints within scheduling optimisation, future research should develop dedicated DSSs for these domains. This includes predictive maintenance approaches that account for uncertainty and degradation dynamics, quality decision support that integrates causal reasoning with process-level monitoring, and cross-level frameworks that connect process-level anomaly detection with factory-level planning. Combining these domains with real- time data and DT environments could lead to DSSs that are more comprehensive and practically relevant for HMLV manufacturing. Practical barriers include limited failure and defect data, uncertain degradation behaviour, heterogeneous sensor data, and the need to coordinate production, maintenance, and quality personnel. Initial studies could focus on a single asset or production process, using existing condition-monitoring or inspection data and allowing human approval of recommendations. Their impact could be evaluated through avoided downtime, reduced scrap, and improved delivery performance. This direction has substantial industrial potential in HMLV manufacturing, but its feasibility depends on the availability and quality of operational data.

**Cross-level integration and broadening beyond scheduling.** The reviewed literature is heavily segmented by manufacturing level, with very few approaches connecting decisions across levels. Among the 205 papers reviewed, only isolated examples bridge manufacturing levels, such as [156, 157, 159, 162]. [156] uses a hierarchical multi-agent RL approach to coordinate distributed scheduling across factories (L3) while generating schedules at the factory level (L2). The near-

absence of such cross-level approaches is a notable gap, as in real HMLV manufacturing, decisions at different levels are interdependent. Because HMLV runs in small batches, a decision optimised in isolation, for instance a schedule that ignores process-level setup and energy conditions or enterprise-level supply constraints, carries more consequence than in high-volume production, where the same cost is absorbed across many units. Coordinating decisions across levels is therefore likely to yield larger practical gains in HMLV than pursuing each level in isolation. Future research should investigate how process-level insights, such as quality feedback or energy consumption patterns, can inform factory level scheduling, and how decisions across levels can be coherently integrated and jointly optimised. More broadly, giving greater attention to logistics, plant reconfiguration, forecasting, and other decision domains beyond scheduling would better reflect the full range of operational challenges in HMLV manufacturing. This broadening matters because the least-studied domains, quality and maintenance (2.9%) and logistics and supply chain (6.3%), are precisely those where a single defect or equipment failure represents a disproportionate share of output in low-volume production. Widening the research focus beyond scheduling, which already accounts for 69.8% of the corpus, is therefore likely to deliver greater practical benefit to HMLV manufacturing than further refinement of scheduling methods alone.

Taken together, these directions should be pursued in stages. Industrial validation is the highest immediate priority because it establishes whether the proposed approaches work under realistic conditions. Human involvement should be addressed alongside validation to ensure that the approaches are usable and acceptable in practice. Infrastructure-integrated DTs are a medium-term enabler, while maintenance and quality applications can progress in parallel wherever sufficient operational data are available.

# 7 Conclusion

This survey has reviewed 205 papers on decision support in HMLV manufacturing, structured across four manufacturing levels and analysed through four research questions.

Regarding **RQ1** (drivers), the development of decision support is driven by the combinatorial complexity and variability inherent in HMLV production. At the process level, energy cost and changeover frequency are the primary drivers. At the line and cell level, the need to coordinate multiple resources including humans and robots under varying product configurations takes over. At the factory level, scheduling complexity under dynamic conditions dominates. At the enterprise level, distributed coordination and supply chain resilience are the key concerns.

Regarding **RQ2** (objectives), makespan minimisation is the dominant objective across all levels, particularly at the factory level. Multi-objective formulations combining makespan with tardiness, energy consumption, or cost are growing, especially at the factory and enterprise levels. However, objectives related to product quality, human workload, and system-level resilience remain underrepresented relative to their importance in HMLV environments.

Regarding **RQ3** (technologies and methods), there is a clear progression across manufacturing levels. Metaheuristics dominate at the process level, adaptive methods such as RL and MCTS gain prominence at the line and cell level, and RL with GNNs becomes the dominant approach at the factory level. At the enterprise level, the landscape is more mixed, with RL, ML, and DTs each serving different roles. Across all levels, integration with enabling infrastructure such as DTs, IoT,

and MESs remains limited.

Regarding **RQ4** (evaluation), within the reviewed corpus, the vast majority of studies rely on simulation-based or static evaluation settings. Only 5 out of 205 papers present evaluation of their techniques in a real industrial setting, which indicates a large gap between methodological progress and real-world validation. Although industrial data is used in 31.7% of studies, deployment and evaluation within operational production environments is rare. However, as discussed in subsection 7.1, the literature collection process may have under-represented industry-specific research and, consequently, studies involving industrial evaluation.

Looking ahead, the field is growing rapidly, with publications increasing from 45 in 2024 to 81 in 2025, supported by the maturation of enabling technologies. However, this growth has been concentrated in scheduling at the factory level. The main challenge for the next phase of research is not to develop better algorithms for well-defined benchmark problems, where the field has already made good progress, but to close the gap between these algorithmic advances and the real manufacturing environments they are intended to serve. This means moving decision support out of simulation and into production, integrating it with the digital infrastructure of the factory, and designing systems that work with human operators rather than replacing them.

### 7.1 Limitations

Survey studies typically involve some degree of human judgement, particularly when defining classification schemes and categorising papers [203]. In this review, papers were classified across manufacturing levels, decision areas, technologies, objectives, and evaluation settings, and some papers naturally spanned multiple categories. In addition, the use of Scopus only, the English-language restriction, and the technically oriented rule-based screening/filtering process may have introduced selection bias. Consequently, studies using different terminology or focusing on less technical aspects of HMLV manufacturing may be underrepresented. Retrospectively, the analysis indicates that a large proportion of the selected papers focus on L2 decision support in HMLV manufacturing. This distribution may be influenced by the design of the Scopus search query, including the use of "job shop". The filtering process may also have contributed to this tendency, as relevance was partly defined through the presence of technological indicators and decision-support-enabling content. As a result, papers in the literature focusing on managerial, organisational, business, policy, or social aspects may be less represented.

## Conflict of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Author Contributions

Petersen, M.E.S: Conceptualisation, Methodology, Software, Formal analysis, Data curation, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualisation. Birkebæk, M.T: Methodology, Software, Formal analysis, Data curation, Validation, Investigation, Writing –

original draft, Writing – review & editing, Visualisation. Braas, M: Formal analysis, Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. Dzigurski, F: Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. Fibiger, J: Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. Lunding, R.S: Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. Yousaf, S: Investigation, Data curation, Validation, Writing – original draft, Writing – review & editing. Sørensen, J.C: Funding acquisition, Writing – original draft, Writing – review & editing. Schou, C: Funding acquisition, Writing – original draft, Writing – review & editing. Larsen, P.G: Supervision, Project administration, Funding acquisition, Writing – original draft, Writing – review & editing. Esterle, L: Supervision, Project administration, Funding acquisition, Writing – original draft, Writing – review & editing.


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## Data Availability Statement

All data collected and produced in this survey are presented on an interactive page: `https://hmlv-survey.vercel.app/`. Scripts and raw data files are available at `https://github.com/MartinBraas-AU/HMLV-survey`. The raw data are also archived on Zenodo at https://doi.org/10.5281/zenodo.22304744.

# A Appendix

## A.1 Relevance Script filter description and criteria

In this section, the relevance script data processing steps are described. Describing which data is used and how it is processed to assign a relevance score to each of the papers identified by the Scopus initial search, i.e. before manually screening and after initial removal of duplicates.

### A.1.1 Text Aggregation

For each publication, the script extracts the *Title*, *Abstract*, *Author Keywords*, and *Indexed Keywords*. These fields are concatenated into a single text string and converted to lowercase to ensure consistent matching. This consolidated representation forms the basis for keyword detection and scoring.

### A.1.2 Keyword Detection and Matching

The relevance script relies on a set of curated regular expression groups, each representing a conceptual category relevant to decision support and Industry 4.0 in the manufacturing industry. Types of keywords include:

- **Core Industry 4.0 technologies:** AI, DT, IoT/Industrial Internet of Things (IIoT), robotics, machine learning architectures, autonomous systems, predictive maintenance, cyber-physical systems, and more.
- **Computational and analytical methods:** ML algorithms, optimisation techniques, RL algorithms, clustering, regression, simulation modelling, finite element methods, and data-driven analytics.
- **Manufacturing/production concepts:** terms indicating production or factory settings, such as job shop, flow shop, assembly lines, HMLV, shop-floor control, industrial automation, and production-(planning, schedule, line, system).
- **Decision support concepts:** DSSs, MCDM, planning support, and intelligent or data-driven decision-making frameworks.
- **Scheduling, metaheuristics, simulation-based methods, robotics, and monitoring related terms**: Technologies enabling decision support.
- **Broad Industry terms:** industry 4.0/5.0, smart factory, digital manufacturing, smart production, digital transformation, cloud computing, and big data.

Based on the aggregated information of individual papers and the regular expression containing the above-described categories, the number of appearances (i.e. keyword matches) is counted. Before classification, the script screens for non-technical indications such as management, policy, business models, HR, finance, or social impact topics. Papers which contain the non-technical terms defined in the relevance script are tagged. To highlight potential non-relevant papers in the review process that discuss Industry 4.0 only from a managerial or policy-oriented perspective.

### A.1.3 Relevance Classification Rule-based Logic

Each paper is then evaluated through a rule-based decision tree based on its match counts.

A paper is classified as **"++" most Relevant** when it:

- Contains technical content (technology match count $\geq 2$).
- Is situated within a manufacturing context (manufacturing match count $\geq 1$).
- Demonstrates a contribution to decision support through:
  - explicit decision support terminology (decision match count $\geq 1$), or
  - Decision support enabling content such as scheduling, metaheuristics, simulation, robotics, and monitoring (combined operational count $\geq 3$).

This level represents papers central to technical decision support in HMLV Industry 4.0 environments.

A paper is classified as **"+" Relevant** when it:

- Contains multiple technological indicators (tech count $\geq 2$).
- Identical to most relevant, includes enabling content (combined count $\geq 2$),
- Appears to be within a manufacturing context.

Alternatively, papers with strong Industry 4.0 contextual indicators (industry count $\geq 2$) and moderate technological content may also receive this classification.

A paper is considered **"*" Possibly Relevant** when:

- It includes at least one technological indicator.
- Is situated within a manufacturing setting.
- Lacks decision support, optimisation, or computational content.

A paper is **"–" Not Relevant**, i.e. to be excluded, when it:

- Lacks manufacturing relevance,
- Focuses on managerial, business, policy, or social aspects,
- Does not contribute technological or methodological elements related to decision support.

## A.2 Benchmark datasets and publications

In this section, all public benchmark datasets and the corresponding publications using them are presented. See Table 9.

Table 9: Public benchmark datasets used in factory level and enterprise and supply chain level papers. Adapted and extended benchmarks are excluded and discussed separately.

| **Benchmark dataset** | **Typical problem context** | **Year** | **Original benchmark reference** | **Notes** | **Papers using the benchmark** |
|---|---|---|---|---|---|
| Brandimarte (mk) | FJSP | 1993 | [139] | Classic flexible job shop benchmark. | [55, 96–99, 111, 113, 114, 136, 186, 191, 204–209] |
| Fisher-Thompson (ft) | JSSP | 1963 | [140] | Standard JSSP benchmark. | [115, 116, 187, 210–217] |
| Lawrence (la) | JSSP | 1984 | [141] | Appears also in derived variants (e.g., via Hurink). | [99, 115, 116, 187, 191, 205, 211–214, 216, 217] |
| Applegate-Cook (orb) | JSSP | 1991 | [142] | Standard JSSP benchmark. | [115, 116, 211–214, 216, 217] |
| Adams-Balas-Zawack (abz) | JSSP | 1988 | [143] | Standard JSSP benchmark. | [213, 214, 216, 217] |

| | | | | | |
|---|---|---|---|---|---|
| Taillard (ta) | JSSP | 1993 | [144] | Widely used benchmark set. | [157, 205, 213, 214, 217–219] |
| Demirkol-Metha-Uzsoy (dmu) | JSSP | 1998 | [145] | Extended JSSP benchmark instances. | [136, 205, 214, 217, 220] |
| Storer-Wu-Vaccari (swv) | JSSP | 1992 | [146] | Standard JSSP benchmark. | [187, 214, 217] |
| Yamada-Nakano (yn) | JSSP | 1992 | [147] | Less frequently used benchmark. | [214] |
| Hurink (edata/rdata/-vdata/sdata) | FJSP | 1994 | [148] | Derived flexible job shop benchmarks. | [96, 98, 99, 108, 111, 112, 116, 136, 191, 206, 208, 221] |
| Dauzère | FJSP | 1997 | [149] | Flexible job shop benchmark. | [55, 96, 114, 205, 206] |
| Kacem | FJSP | 2002 | [150] | Widely used FJSP benchmark. | [55, 128, 222] |
| Fattahi / Homayouni-Fontes | FJSP | 2007 | [151] | Flexible job shop benchmark. | [122, 223] |
| Behnke | FJSP | 2014 | [152] | Less standardized benchmark. | [108, 221] |
| Deroussi-Norre | FJSP | 2010 | [153] | Specialized benchmark (AGV context). | [112] |
| Boccia | AGV scheduling problem | 2023 | [154] | Limited context information. | [130] |

# References


[1] Z. L. Gan, S. N. Musa, and H. J. Yap, "A review of the high-mix, low-volume manufacturing industry," Applied Sciences, vol. 13, no. 3, 2023.

[2] H. ElMaraghy, G. Schuh, W. ElMaraghy, F. Piller, P. Schönsleben, M. Tseng, and A. Bernard, "Product variety management," CIRP Annals, vol. 62, no. 2, pp. 629–652, 2013.

[3] H. Sheng, T. Feng, L. Chen, and D. Chu, "Responding to market turbulence by big data analytics and mass customization capability," Industrial Management and Data Systems, 2021.

[4] J. Choi and S. B. Kim, "Deep reinforcement learning for scheduling semiconductor cluster tools in varying configurations," Scientific Reports, vol. 16, no. 1, 2025. [Online]. Available: http://dx.doi.org/10.1038/s41598-025-31722-7

[5] T. Tahmina, M. Garcia, Z. Geng, and B. Bidanda, "A survey of smart manufacturing for high-mix low-volume production in defense and aerospace industries," in Proceedings of the International Conference on Flexible Automation and Intelligent Manufacturing: The Human-Data-Technology Nexus, K.-Y. Kim, L. Monplaisir, and J. Rickli, Eds. Cham: Springer International Publishing, 2023, pp. 237–245.

[6] S. Bednar and E. Rauch, "Modeling and application of configuration complexity scale: concept for customized production," International Journal of Advanced Manufacturing Technology, 2019.

[7] C. Y. Long, S. B. Khamis, J. Lim, and T. C. Ee, "Real-time decision-support system for high-mix low-volume production scheduling in industry 4.0," 2025.

[8] K. Johansen, S. Rao, and M. Ashourpour, "The role of automation in complexities of high-mix in low-volume production – a literature review," Procedia CIRP, vol. 104, pp. 1452–1457, 2021, 54th CIRP CMS 2021 - Towards Digitalized Manufacturing 4.0.

[9] B. Scholten and D. L. Brandl, The road to integration: A guide to applying the ISA-95 Standards in Manufacturing. John Wiley & Sons, 2022.

[10] International Organization for Standardization, "IEC 62264-1:2013: Enterprise-control system integration - part 1: Models and terminology," International Organization for Standardization, Tech. Rep. ISO 62264-1:2013, 2013. [Online]. Available: https://www.iso.org/standard/57308.html

[11] M. Hankel and B. Rexroth, "The reference architectural model industrie 4.0 (rami 4.0)," Zvei, vol. 2, no. 2, pp. 4–9, 2015.

[12] H. Kagermann, W. Wahlster, and J. Helbig, "Recommendations for implementing the strategic initiative INDUSTRIE 4.0: Final report of the Industrie 4.0 working group," acatech – National Academy of Science and Engineering, Tech. Rep., 2013.

[13] F. M. Kasie, G. Bright, and A. Walker, "Decision support systems in manufacturing: a survey and future trends," Journal of Modelling in Management, vol. 12, no. 3, pp. 432–454, 2017.

[14] E. Mahmoodi, M. Fathi, M. Tavana, M. Ghobakhloo, and A. H. Ng, "Data-driven simulation-based decision support system for resource allocation in industry 4.0 and smart manufacturing," Journal of Manufacturing Systems, 2024, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[15] Y. Wang, H. Liu, P. Zheng, C. Ye, and S. Li, "A literature review of reinforcement learning methods applied to job-shop scheduling problems," Computers & Operations Research, vol. 174, p. 106929, 2025.

[16] C. Ngwu, Y. Liu, and R. Wu, "Reinforcement learning in dynamic job shop scheduling: a comprehensive review of ai-driven approaches in modern manufacturing," Journal of Intelligent Manufacturing, 2025, all Open Access; Hybrid Gold Open Access.

[17] T. Tahmina, M. Garcia, Z. Geng, and B. Bidanda, "A survey of smart manufacturing for high-mix low-volume production in defense and aerospace industries," Flexible Automation and Intelligent Manufacturing: The Human-Data-Technology Nexus, pp. 237–245, 2023.

[18] Z. Chen and R. Zhang, "Deep reinforcement learning for job shop scheduling problems: A comprehensive literature review," Knowledge-Based Systems, vol. 312, p. 113633, 2025.

[19] M. Dalle Mura and G. Dini, "Graph neural networks for job shop scheduling problems: A survey," Computers & Operations Research, vol. 176, p. 106914, 2025.

[20] P. Meilanitasari and S.-J. Shin, "A review of prediction and optimization for sequence-driven scheduling in job shop flexible manufacturing systems," Processes, vol. 9, no. 8, 2021.

[21] T. Böttjer, D. Tola, F. Kakavandi, C. R. Wewer, D. Ramanujan, C. Gomes, P. G. Larsen, and A. Iosifidis, "A review of unit level digital twin applications in the manufacturing industry," CIRP Journal of Manufacturing Science and Technology, vol. 45, pp. 162–189, 2023.

[22] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan, R. Chou, J. Glanville, J. M. Grimshaw, A. Hróbjartsson, M. M. Lalu, T. Li, E. W. Loder, E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, J. Thomas, A. C. Tricco, V. A. Welch, P. Whiting, and D. Moher, "The prisma 2020 statement: an updated guideline for reporting systematic reviews," BMJ, vol. 372, 2021.

[23] D. Feldner, "Scopus data crosses the 100 million item threshold!" Elsevier Scopus Blog. Available at: https://blog.scopus.com/100-million-reasons-to-trust-scopus/, 2025, published: 2025-03-03; Accessed: 2026-01-01.

[24] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, Experimentation in Software Engineering, 2024, cited by: 63; All Open Access, Green Open Access.

[25] A. M. Law, Simulation Modeling and Analysis, 5th ed. McGraw-Hill Education, 2015.

[26] H. H. Hoos and T. Stützle, Stochastic Local Search: Foundations and Applications. Elsevier / Morgan Kaufmann, 2005.

[27] E.-G. Talbi, Metaheuristics: From Design to Implementation. Wiley, 2009.

[28] S. S. A. Rubaiee, S. Çınar, and M. B. Yildirim, "An energy-aware multiobjective optimization framework to minimize total tardiness and energy cost on a single-machine nonpreemptive scheduling," IEEE Transactions on Engineering Management, 2019, all Open Access; Green Accepted Open Access; Green Open Access.

[29] Y.-C. Tsao, V.-V. Thanh, and F.-J. Hwang, "Energy-efficient single-machine scheduling problem with controllable job processing times under differential electricity pricing," Resources, Conservation and Recycling, vol. 161, p. 104902, 2020.

[30] S. Lee, M. Issabakhsh, H. W. Jeon, S. W. Hwang, and B. D. Chung, "Idle time and capacity control for a single machine scheduling problem with dynamic electricity pricing," Operations Management Research, vol. 13, no. 3, pp. 197–217, 2020.

[31] A. Che, Y. Zeng, and K. Lyu, "An efficient greedy insertion heuristic for energy-conscious single machine scheduling problem under time-of-use electricity tariffs," Journal of Cleaner Production, vol. 129, pp. 565–577, 2016.

[32] P. Wu, N. Li, J. Cheng, and C. Chu, "Energy-efficient single-machine scheduling with release dates under time-of-use electricity tariffs," Journal of Cleaner Production, vol. 393, p. 136228, 2023.

[33] L. Xue and X. Wang, "A multi-objective artificial bee colony algorithm for single machine scheduling with family setup under tou tariffs," International Journal of Production Research, vol. 63, no. 10, pp. 3822–3853, 2025.

[34] M. L. Bittencourt, C. M. R. de Oliveira, I. D. Lins, and R. Kramer, "A simheuristic approach to optimize energy consumption in the single-machine scheduling problem with stochastic processing times," Computers and Industrial Engineering, vol. 210, p. 111580, 2025.

[35] C. Li, P. Zheng, P. Zhou, Y. Yin, C. K. M. Lee, and L. Wang, "Unleashing mixed-reality capability in deep reinforcement learning-based robot motion generation towards safe human–robot collaboration," Journal of Manufacturing Systems, 2024.

[36] C. Zhang Sprenger, J. A. Corrales Ramón, and N. U. Baier, "Rtmn 2.0—an extension of robot task modeling and notation (rtmn) focused on human–robot collaboration," Applied Sciences (Switzerland), 2024, all Open Access; Gold Open Access.

[37] T. Hoebert, W. Lepuschitz, M. Vincze, and M. Merdan, "Knowledge-driven framework for industrial robotic systems," Journal of Intelligent Manufacturing, 2023.

[38] A. Devanga, E. D. Badilla, and M. Dehghanimohammadabadi, "Applied reinforcement learning for decision making in industrial simulation environments," 2022.

[39] E. Yu, Z. Yu, L. Hu, M. Hu, Y. Lu, W. Gao, J. Li, and J. Liu, "Design and optimization of smt reflow soldering curves driven by digital twins," 2024.

[40] B. Tanane, B. Dafflon, M. L. Bentaha, N. Moalla, and V. Ferreiro, "Toward a collaborative sensor network integration for smes' zero-defect manufacturing," IFIP Advances in Information and Communication Technology, 2022, all Open Access; Green Accepted Open Access; Green Open Access.

[41] M. Slot, E. Navis, R. Damgrave, and E. Lutters, "Versatile information provisioning in a configure-to-order production environment; a case study," 2023, all Open Access; Gold Open Access.

[42] Q. Wang, C. Zhang, Y. Ding, G. Xydis, J. Wang, and J. Østergaard, "Review of real-time electricity markets for integrating distributed energy resources and demand response," Applied Energy, vol. 138, pp. 695–706, 2015.

[43] J. Ahn and H. J. Kim, "A novel mixed integer programming model with precedence relation-based decision variables for non-cyclic scheduling of cluster tools," IEEE Transactions on Automation Science and Engineering, 2025.

[44] H. J. Kim and J. Lee, "Deep reinforcement learning with a look-ahead search for robotic cell scheduling," IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024.

[45] Y. Du, Y. Wang, J. Wang, and T. Zhang, "Production task allocation decision based on cloud robot cell-line," Computational Intelligence and Neuroscience, 2022, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[46] H. Guo, K. Li, J. Liu, C. Zhuang, and F. Pei, "Dynamic integrated process planning and scheduling under multi-resource constraints in workshops with reconfigurable manufacturing cells: a novel hyper-heuristic approach," Expert Systems with Applications, vol. 289, p. 128337, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.eswa.2025.128337

[47] B. Tian, H. Kaul, and M. Janardhanan, "Balancing heterogeneous assembly line with multi-skilled human-robot collaboration via adaptive cooperative co-evolutionary algorithm," Swarm and Evolutionary Computation, 2024, all Open Access; Hybrid Gold Open Access.

[48] R. Senington, B. Schmidt, and A. Syberfeldt, "Monte carlo tree search for online decision making in smart industrial production," Computers in Industry, 2021, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[49] J. Zhang and S. Fujimura, "An innovative meta–heuristic for balancing and scheduling human–robot collaborative assembly lines in industry 5.0," Journal of Industrial and Production Engineering, vol. 43, no. 1, p. 117–137, 2025. [Online]. Available: http://dx.doi.org/10.1080/21681015.2025.2533436

[50] C. Urrea, "Hybrid deep learning-reinforcement learning for adaptive human-robot task allocation in industry 5.0," Systems, vol. 13, no. 8, p. 631, 2025. [Online]. Available: http://dx.doi.org/10.3390/systems13080631

[51] E. Mahmoodi, M. Fathi, A. H. Ng, and A. Dolgui, "Predictive model-based multi-objective optimization with life-long meta-learning for designing unreliable production systems," Computers &; Operations Research, vol. 178, p. 107011, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.cor.2025.107011

[52] A. Kochański, J. Kozłowski, M. Perzyk, and H. Sadłowska, "Data-driven advisory system for industrial manufacturing. application to the aluminum extrusion process," Knowledge-Based Systems, vol. 294, p. 111631, 2024. [Online]. Available: http://dx.doi.org/10.1016/j.knosys.2024.111631

[53] S. Fu, X. Yue, Q. Ma, X. Xiong, Q. Lin, and W. Lin, "A hybrid optimization approach integrating genetic algorithm and monte carlo tree search for multi-machine scheduling in manufacturing irregular solid wood furniture parts," Wood Material Science & Engineering, pp. 1–14, 2025. [Online]. Available: http://dx.doi.org/10.1080/17480272.2025.2505133

[54] H. Li, Y. Cao, Y. Lei, H. Cao, J. Peng, and Y. Jia, "Energy-aware dynamic rescheduling of flexible manufacturing system using edge-cloud collaborative decision-making method," International Journal of Computer Integrated Manufacturing, 2025.

[55] M. Liu, Z. Rui, X. Zhang, M. G. Ge, L. Ling, X. Wang, and C. Liu, "Efjsp-ibdr: Energy-efficient flexible job shop scheduling under incentive-based demand response via graph reinforcement learning with dual attention mechanism," Expert Systems with Applications, 2025.

[56] G. Xu, Q. Bao, and H. Zhang, "Multi-objective green scheduling of integrated flexible job shop and automated guided vehicles," Engineering Applications of Artificial Intelligence, 2023.

[57] E. Yuan, L. Wang, S. Song, S. Cheng, and W. Fan, "Dynamic scheduling for multi-objective flexible job shop via deep reinforcement learning," Applied Soft Computing, 2025.

[58] R. Duan, S. Wang, Y. Liu, W. Yan, Z. Jiang, and Z. Pan, "A multi-trigger mechanism design for rescheduling decision assistance in smart job shops based on machine learning," Sustainability (Switzerland), 2025, all Open Access; Gold Open Access.

[59] M. Klar, M. Hussong, P. Flore, L. Yi, M. F. Glatt, and J. C. Aurich, "Scalability investigation of double deep q learning for factory layout planning," 2022, all Open Access; Gold Open Access.

[60] M. Li, T. Qu, B. Liu, Q. Luo, M. Yan, M. Li, Z. He, and G. Q. Huang, "Out-of-order architecture for real-time data-driven resilient planning and scheduling of cyber-physical manufacturing systems," IEEE Transactions on Automation Science and Engineering, vol. 23, pp. 2983–2994, 2026. [Online]. Available: http://dx.doi.org/10.1109/TASE.2025.3563084

[61] G. Fragapane, D. Ivanov, M. Peron, F. Sgarbossa, and J. O. Strandhagen, "Increasing flexibility and productivity in industry 4.0 production networks with autonomous mobile robots and smart intralogistics," Annals of Operations Research, vol. 308, no. 1–2, p. 125–143, 2020. [Online]. Available: http://dx.doi.org/10.1007/s10479-020-03526-7

[62] K. Mahmood, K. Karjust, and T. Raamets, "Production intralogistics automation based on 3d simulation analysis," Journal of Machine Engineering, p. 101–115, 2021. [Online]. Available: http://dx.doi.org/10.36897/jme/137081

[63] J. Lee, P. C. Chua, B. Liu, S. K. Moon, and M. Lopez, "A hybrid data-driven optimization and decision-making approach for a digital twin environment: Towards customizing production platforms," International Journal of Production Economics, 2025.

[64] X. Feng, Z. Zhao, and C. Zhang, "Simulation optimization framework for online deployment and adjustment of reconfigurable machines in job shops," 2020.

[65] C. Baltazar, J. C. Reis, and G. M. Gonçalves, "Prescriptive system for reconfigurable manufacturing systems considering variable demand and production rates," 2020.

[66] M. B. Canciglieri, E. D. F. Rocha Loures, A. L. Szejka, Y. M. Goh, and R. P. Monfared, "Smart reconfigurable manufacturing system based on semantic models and multicriteria decision making," IEEE Access, 2025, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[67] G. Lefranc, I. LOPEZ-JUAREZ, and G. GATICA, "Enhancing fms performance through multi-agent systems in the context of industry 4.0," Studies in Informatics and Control, vol. 33, no. 2, p. 5–14, 2024. [Online]. Available: http://dx.doi.org/10.24846/V33I2Y202401

[68] I. Gödri, C. Kardos, A. Pfeiffer, and J. Vancza, "Data analytics-based decision support workflow for high-mix low-volume production systems," CIRP Annals, 2019.

[69] J. Flyckt and N. Lavesson, "Navigating demand forecasting in make-to-order manufacturing: the role of global models and intermittent time-series," 2025.

[70] C. Li and Q. Chang, "Generative ai-powered planning: A hybrid graph-diffusion approach for demand-driven flexible manufacturing systems," Journal of Manufacturing Systems, vol. 83, pp. 175–195, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.jmsy.2025.08.016

[71] M. C. Herkes and G. Oversluizen, "Using a systems approach to model a process digital twin," 2022, all Open Access; Gold Open Access.

[72] P. H. Modesti, J. Kobylarz, and M. Borsato, "Artificial intelligence-based method for forecasting flowtime in job shops," VINE Journal of Information and Knowledge Management Systems, 2024.

[73] W. Cen, C. Su, K. Cen, L. Yang, G. Chen, and L. Xie, "A dual spatial temporal neural network for bottleneck prediction in manufacturing systems," Engineering Applications of Artificial Intelligence, vol. 159, p. 111586, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.engappai.2025.111586

[74] A. Chawalitanont, A. Bashyal, and H. A. Wicaksono, "Uncertainty-aware power consumption prediction in customized stainless-steel manufacturing: A comparative study of hierarchical bayesian and deep neural models," Journal of Manufacturing Systems, 2025, all Open Access; Hybrid Gold Open Access.

[75] M. F. Glatt, C. Sinnwell, L. Yi, S. P. Donohoe, B. Ravani, and J. C. Aurich, "Modeling and implementation of a digital twin of material flows based on physics simulation," Journal of Manufacturing Systems, 2021, all Open Access; Hybrid Gold Open Access.

[76] M. Elbasheer, V. D'Augusta, G. Mirabelli, V. Solina, and S. Talarico, "Leveraging auto-generative simulation for decision support in engineer-to-order manufacturing," 2024, all Open Access; Gold Open Access.

[77] M. Elbasheer, Y. Laili, F. Longo, V. Solina, Y. Tao, P. Veltri, Y. Zhang, and L. Zhang, "Natural language-driven production planning: integrating large language models with automatic simulation model generation in manufacturing systems," Journal of Intelligent Manufacturing, 2025. [Online]. Available: http://dx.doi.org/10.1007/s10845-025-02732-z

[78] T. H. Kim, B. I. Gu, K. H. Kwon, and A.-N. Kim, "A framework for 3d plant simulation of meal-kit-packaging robot automation system," Applied Sciences, vol. 15, no. 8, p. 4116, 2025. [Online]. Available: http://dx.doi.org/10.3390/app15084116

[79] T. Coito, M. S. Martins, J. L. Viegas, B. M. Firme, J. M. G. Figueiredo, S. M. Vieira, and J. M. da Costa Sousa, "A middleware platform for intelligent automation: An industrial prototype implementation," Computers in Industry, 2020, all Open Access; Bronze Open Access; Green Accepted Open Access; Green Open Access.

[80] T. Jiang, J. Zhou, J. Zhao, M. Wang, and S. Zhang, "A multi-dimensional cognitive framework for cognitive manufacturing based on oar model," Journal of Manufacturing Systems, 2022.

[81] W. Fang, Y. Guo, W. Liao, S. Huang, C. Yang, and K. Cui, "The spatio-temporal modeling and integration of manufacturing big data in job shop: An ontology-based approach," 2020.

[82] J. Yang, X. Lan, Z. Yao, W. Liu, W. Deng, and W. Qu, "Research on production process control method of discrete manufacturing digital workshop," 2022.

[83] M. Feng and Y. Li, "Predictive maintenance decision making based on reinforcement learning in multistage production systems," IEEE Access, 2022, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[84] H. Wang, Z. Zhang, X. Wang, Z. Ye, X. Cui, and Z. Cai, "Joint optimization of maintenance and quality inspection for multi-stage manufacturing system based on genetic reinforcement learning," Quality and Reliability Engineering International, vol. 41, no. 7, p. 2879–2896, 2025. [Online]. Available: http://dx.doi.org/10.1002/qre.70027

[85] X. He, Z. Wang, Y. Li, S. Khazhina, W. Du, J. Wang, and W. Wang, "Joint decision-making of parallel machine scheduling restricted in job-machine release time and preventive maintenance with remaining useful life constraints," Reliability Engineering and System Safety, 2022.

[86] R. H. Bemthuis, M. Koot, M. R. Mes, F. A. Bukhsh, M. E. Iacob, and N. Meratnia, "An agent-based process mining architecture for emergent behavior analysis," 2019, all Open Access; Green Accepted Open Access; Green Open Access.

[87] L. Ling, Z. M. Song, X. Zhang, P. Z. Cao, X. Wang, C. Liu, and M. Liu, "Manufacturing task data chain-driven production logistics trajectory analysis and optimization decision making method," Advances in Manufacturing, 2024.

[88] O. Antons and J. C. Arlinghaus, "Designing distributed decision-making authorities for smart factories–understanding the role of manufacturing network architecture," International Journal of Production Research, 2024.

[89] S. Skėrė, A. Žvironienė, K. Juzėnas, and S. Petraitiene, "Optimization experiment of production processes using a dynamic decision support method: A solution to complex problems in industrial manufacturing for small and medium-sized enterprises," Sensors, 2023, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[90] R. Hou, S. Xu, C. Yang, J. Duan, X. Wu, and T. Xu, "A learning-based assembly sequence planning method using neural combinatorial optimization with satisfactory generalization ability," IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 8952–8964, 2025. [Online]. Available: http://dx.doi.org/10.1109/TASE.2024.3493617

[91] Z. He, M. Thürer, and W. Zhou, "The use of reinforcement learning for material flow control: An assessment by simulation," International Journal of Production Economics, 2024, all Open Access; Hybrid Gold Open Access.

[92] P. Rubel, W. Motsch, A. T. Bernhard, S. Jungbluth, and M. Ruskowski, "Agent-based communication for fault diagnosis in skill-based production environments using messages based on i4.0 language and asset administration shells," Lecture Notes in Mechanical Engineering, 2025, all Open Access; Hybrid Gold Open Access.

[93] M. Sanowar Hossain, R. K. Chakrabortty, S. El Sawah, and M. J. Ryan, "A multi-objective bi-level leader-follower joint optimization for concurrent design of product family and assembly system," Computers and Industrial Engineering, vol. 177, p. 109035, 2023. [Online]. Available: http://dx.doi.org/10.1016/j.cie.2023.109035

[94] A. Ghasemi, K. E. Kabak, and C. Heavey, "Demonstration of the feasibility of real time application of machine learning to production scheduling," 2022.

[95] M.-s. Kim, S. Oh, E. Chang, S. Lee, J. W. Wells, J. F. Arinez, and Y. J. Jang, "A dynamic programming-based heuristic algorithm for a flexible job shop scheduling problem of a matrix system in automotive industry," 2022.

[96] Z. Zhang, T. Huang, B. Qian, and R. Hu, "Mamhsan: A multi-agent deep reinforcement learning framework based on multi-head self-attention network with heterogeneous graph embedding for flexible job shop scheduling," Computers and Industrial Engineering, 2025.

[97] S. Moon, S. Lee, and K. J. Park, "Learning-enabled flexible job-shop scheduling for scalable smart manufacturing," Journal of Manufacturing Systems, 2024.

[98] R. Wang, G. Wang, J. Sun, F. Deng, and J. Chen, "Flexible job shop scheduling via dual attention network-based reinforcement learning," IEEE Transactions on Neural Networks and Learning Systems, 2024.

[99] W. Song, X. Chen, Q. Li, and Z. Cao, "Flexible job-shop scheduling via graph neural network and deep reinforcement learning," IEEE Transactions on Industrial Informatics, 2023.

[100] R. Wang, C. Liu, X. Wang, S. Yang, and Y. Hou, "A multi-action deep reinforcement learning based on bilstm for flexible job shop scheduling problem with tight time," 2025, all Open Access; Gold Open Access.

[101] Z. Wang and W. Liao, "Smart scheduling of dynamic job shop based on discrete event simulation and deep reinforcement learning," Journal of Intelligent Manufacturing, 2024.

[102] Z. Hou, L. Zhang, Y. Wang, and Y. Hu, "Deep reinforcement learning for dynamic flexible job-shop scheduling with automated guided vehicles," Lecture Notes in Mechanical Engineering, 2024.

[103] L. Yue, K. Peng, L. Ding, J. Mumtaz, L. Lin, and T. Zou, "Two-stage double deep q-network algorithm considering external non-dominant set for multi-objective dynamic flexible job shop scheduling problems," Swarm and Evolutionary Computation, 2024.

[104] C. Li, K. Bhatta, M. Waseem, and Q. C. Chang, "Demand-driven hierarchical integrated planning-scheduling control for a mobile robot-operated flexible smart manufacturing system," Robotics and Computer-Integrated Manufacturing, 2025.

[105] L. Zhang, Y. Hu, Q. Tang, J. Li, and Z. Li, "Data-driven dispatching rules mining and real-time decision-making methodology in intelligent manufacturing shop floor with uncertainty," Sensors, 2021, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[106] D. Zeng, W. Yang, Y. Du, X. Yang, Y. Wang, and Y. Lv, "Dynamic multi-objective flexible job-shop scheduling via dynamic dual attention network-based deep reinforcement learning," Journal of Supercomputing, 2025.

[107] X. Chen, J. Li, Z. Wang, Q. Chen, K. Gao, and Q. Pan, "Optimizing dynamic flexible job shop scheduling using an evolutionary multitask optimization framework and genetic programming," IEEE Transactions on Evolutionary Computation, 2025.

[108] I. Echeverria, M. Murua, and R. Santana, "Diverse policy generation for the flexible job-shop scheduling problem via deep reinforcement learning with a novel graph representation," Engineering Applications of Artificial Intelligence, 2025, all Open Access; Hybrid Gold Open Access.

[109] J. Petersen, A. Nourmohammadi, M. Fathi, and C. Burmeister, "Multi-objective optimization of transfer line balancing problem considering cycle time and energy expenditure," Procedia CIRP, vol. 130, pp. 1378–1383, 2024. [Online]. Available: http://dx.doi.org/10.1016/j.procir.2024.10.255

[110] Z. Mao, W. Wang, Y. Sun, and K. Fang, "Job shop scheduling problem with a dual-gripper robot," IISE Transactions, pp. 1–16, 2025. [Online]. Available: http://dx.doi.org/10.1080/24725854.2025.2561569

[111] H. Tang and J. Dong, "Solving flexible job-shop scheduling problem with heterogeneous graph neural network based on relation and deep reinforcement learning," Machines, 2024, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[112] X. Zhao, H. Qu, M. Zhang, J. Feng, L. Wang, and Q. Wu, "A flexible job shop scheduling method based on heterogeneous disjunctive graph and deep reinforcement learning," Engineering Applications of Artificial Intelligence, 2025.

[113] J. Du and F. Li, "Graph-augmented deep reinforcement learning for key feature extraction in fjsp," 2025.

[114] A. Corrêa, A. Jesus, C. Silva, P. Peças, and S. Moniz, "Rainbow versus deep q-network: A reinforcement learning comparison on the flexible job-shop problem," 2024, all Open Access; Gold Open Access.

[115] E. K. ElSayed, A. K. Elsayed, and K. A. El-Dahshan, "Deep reinforcement learning-based job shop scheduling of smart manufacturing," Computers, Materials and Continua, 2022, all Open Access; Gold Open Access.

[116] C. Su, C. Zhang, C. Wang, W. Cen, G. Chen, and L. Xie, "Fast pareto set approximation for multi-objective flexible job shop scheduling via parallel preference-conditioned graph reinforcement learning," Swarm and Evolutionary Computation, 2024.

[117] W. Zhang, H. Geng, X. Bao, M. Gen, G. Zhang, and M. Deng, "Expert-guided deep reinforcement learning for flexible job shop scheduling problem," Communications in Computer and Information Science, 2024.

[118] K. Lei, P. Guo, Y. Wang, J. Zhang, X. Meng, and L. Qian, "Large-scale dynamic scheduling for flexible job-shop with random arrivals of new jobs by hierarchical reinforcement learning," IEEE Transactions on Industrial Informatics, 2024.

[119] T. Zhou, D. Tang, H. Zhu, and Z. Zhang, "Multi-agent reinforcement learning for online scheduling in smart factories," Robotics and Computer-Integrated Manufacturing, 2021.

[120] X. Wang, L. Zhang, T. Lin, C. Zhao, K. Wang, and Z. Chen, "Solving job scheduling problems in a resource preemption environment with multi-agent reinforcement learning," Robotics and Computer-Integrated Manufacturing, 2022.

[121] Y. Li, W. B. Gu, M. Yuan, and Y. Tang, "Real-time data-driven dynamic scheduling for flexible job shop with insufficient transportation resources using hybrid deep q network," Robotics and Computer-Integrated Manufacturing, 2022.

[122] C. Li, L. Lin, X. Qiu, X. Zhao, W. Zhang, and M. Gen, "Evolutionary experience-guided deep reinforcement learning for job shop scheduling with agv transportation constraints in flexible manufacturing system," Computers and Industrial Engineering, 2025.

[123] A. Padovano, F. Longo, L. Nicoletti, L. Gazzaneo, A. Chiurco, and S. Talarico, "A prescriptive maintenance system for intelligent production planning and control in a smart cyber-physical production line," 2021, all Open Access; Gold Open Access.

[124] R. F. Santos, H. Piqueiro, R. C. Dias, and C. D. Rocha, "Transitioning trends into action: A simulation-based digital twin architecture for enhanced strategic and operational decision-making," Computers and Industrial Engineering, 2024.

[125] Y. Ma and Y. Shi, "Research on multi-objective scheduling optimization of flexible workshop considering the transportation energy consumption of agv," 2025.

[126] M. Zhang, L. Wang, F. Qiu, and X. Liu, "Dynamic scheduling for flexible job shop with insufficient transportation resources via graph neural network and deep reinforcement learning," Computers and Industrial Engineering, 2023.

[127] Y. Wang, R. Wang, J. Sun, and G. Wang, "Unified drl for enhanced flexible job-shop scheduling with transportation constraints," 2024.

[128] J. Yan, Z. Liu, T. Zhang, and Y. Zhang, "Autonomous decision-making method of transportation process for flexible job shop scheduling problem based on reinforcement learning," 2021.

[129] Y. Li, X. Li, and L. Gao, "Real-time scheduling for production-logistics collaborative environment using multi-agent deep reinforcement learning," Advanced Engineering Informatics, vol. 65, p. 103216, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.aei.2025.103216

[130] S. Zhou, Z. Wang, Y. Li, and X. Wen, "A matheuristic method for the automated guided vehicle scheduling problem with flexible charging and job release," Applied Soft Computing, 2025.

[131] C. Rema, A. J. M. Sousa, H. M. Sobreira, P. Costa, and M. F. F Silva, "Exploring the potential of llm-based chatbots for task scheduling in robot operations," IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC, 2025.

[132] J. Wang, Y. Liu, S. Ren, C. Wang, and W. Wang, "Evolutionary game based real-time scheduling for energy-efficient distributed and flexible job shop," Journal of Cleaner Production, 2021, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[133] X. Wang, Y. Laili, L. Zhang, and Y. Liu, "Hybrid task scheduling in cloud manufacturing with sparse-reward deep reinforcement learning," IEEE Transactions on Automation Science and Engineering, 2025.

[134] A. Kuhnle, L. Schafer, N. Stricker, and G. Lanza, "Design, implementation and evaluation of reinforcement learning for an adaptive order dispatching in job shop manufacturing systems," 2019, all Open Access; Gold Open Access; Green Open Access.

[135] M. A. Ghaleb, S. Taghipour, and H. Zolfagharinia, "Real-time integrated production-scheduling and maintenance-planning in a flexible job shop with machine deterioration and condition-based maintenance," Journal of Manufacturing Systems, 2021.

[136] Y. Pu, F. Li, and S. Rahimifard, "Multi-agent reinforcement learning for job shop scheduling in dynamic environments," Sustainability (Switzerland), 2024, all Open Access; Gold Open Access.

[137] H. Lee and H. Yang, "Cognitive digital twin for multiobjective production scheduling," Journal of Manufacturing Science and Engineering, 2025.

[138] T. Zhou, H. Zhu, D. Tang, C. Liu, Q. Cai, W. Shi, and Y. Gui, "Reinforcement learning for online optimization of job-shop scheduling in a smart manufacturing factory," Advances in Mechanical Engineering, 2022, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[139] P. Brandimarte, "Routing and scheduling in a flexible job shop by tabu search," Annals of Operations Research, vol. 41, no. 3, pp. 157–183, 1993.

[140] H. Fisher and G. L. Thompson, "Probabilistic learning combinations of local job-shop scheduling rules," in Industrial Scheduling, J. F. Muth and G. L. Thompson, Eds. Prentice-Hall, 1963, pp. 225–251.

[141] S. R. Lawrence, "Resource constrained project scheduling: An experimental investigation of heuristic scheduling techniques (supplement)," Ph.D. dissertation, Graduate School of Industrial Administration (GSIA), Carnegie-Mellon University, 1984.

[142] D. L. Applegate and W. J. Cook, "A computational study of the job-shop scheduling problem," ORSA Journal on Computing, vol. 3, no. 2, pp. 149–156, 1991, the JSSP instances used were generated in Bonn in 1986.

[143] J. Adams, E. Balas, and D. Zawack, "The shifting bottleneck procedure for job shop scheduling," Management Science, vol. 34, no. 3, pp. 391–401, 1988.

[144] Éric D. Taillard, "Benchmarks for basic scheduling problems," European Journal of Operational Research, vol. 64, no. 2, pp. 278–285, 1993.

[145] E. Demirkol, S. Mehta, and R. Uzsoy, "Benchmarks for shop scheduling problems," European Journal of Operational Research, vol. 109, no. 1, pp. 137–141, 1998.

[146] R. H. Storer, S. D. Wu, and R. Vaccari, "New search spaces for sequencing problems with application to job shop scheduling," Management Science, vol. 38, no. 10, pp. 1495–1509, 1992.

[147] T. Yamada and R. Nakano, "A genetic algorithm applicable to large-scale job-shop instances," in Proceedings of the International Conference on Parallel Problem Solving from Nature, 1992, pp. 281–290.

[148] J. Hurink, B. Jurisch, and M. Thole, "Tabu search for the job-shop scheduling problem with multi-purpose machines," OR Spektrum, vol. 15, pp. 205–215, 1994.

[149] S. Dauzère-Pérès and J. Paulli, "An integrated approach for modeling and solving the general multiprocessor job-shop scheduling problem using tabu search," Annals of Operations Research, vol. 70, pp. 281–306, 1997.

[150] I. Kacem, S. Hammadi, and P. Borne, "Approach by localization and multiobjective evolutionary optimization for flexible job-shop scheduling problems," IEEE Transactions on Systems, Man, and Cybernetics, Part C, vol. 32, no. 1, pp. 1–13, 2002.

[151] P. Fattahi, M. Saidi-Mehrabad, and R. Tavakkoli-Moghaddam, "Mathematical modeling and heuristic approaches to flexible job shop scheduling problems," Journal of Intelligent Manufacturing, vol. 18, no. 3, pp. 331–342, 2007.

[152] D. Behnke and J. Geiger, "Test instances for the flexible job shop scheduling problem with work centers," Computers & Operations Research, vol. 52, pp. 239–250, 2014.

[153] L. Deroussi and S. Norre, "Simultaneous scheduling of machines and vehicles for the flexible job shop problem," in Proceedings of the International Conference on Metaheuristics and Nature Inspired Computing, 2010, pp. 1–2.

[154] M. Boccia, A. Masone, C. Sterle, and T. Murino, "The parallel agv scheduling problem with battery constraints: A new formulation and a matheuristic approach," European Journal of Operational Research, vol. 307, no. 2, pp. 590–603, 2023.

[155] D. Kopp, M. Hassoun, A. Kalir, and L. Mönch, "Smt2020—a semiconductor manufacturing testbed," IEEE Transactions on Semiconductor Manufacturing, vol. 33, no. 4, pp. 522–531, 2020.

[156] J. Huang, L. Gao, X. Li, and C. J. Zhang, "A cooperative hierarchical deep reinforcement learning based multi-agent method for distributed job shop scheduling problem with random job arrivals," Computers and Industrial Engineering, 2023.

[157] J. Huang, L. Gao, and X. Li, "An end-to-end deep reinforcement learning method based on graph neural network for distributed job-shop scheduling problem," Expert Systems with Applications, 2024.

[158] Y. Du, J.-q. Li, P.-y. Duan, and X.-x. Geng, "A deep reinforcement learning driven bi-population evolutionary optimization for precast concrete scheduling with production transportation," Computers & Industrial Engineering, vol. 207, p. 111354, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.cie.2025.111354

[159] S. Yang, J. Wang, and Z. Xu, "Learning to schedule dynamic distributed reconfigurable workshops using expected deep q-network," Advanced Engineering Informatics, 2024.

[160] K. Zhang, Q. Gao, J. Lü, M. J. Ogorzałek, and Y. Deng, "A subgraph-based hierarchical q-learning approach to optimal resource scheduling for complex industrial networks," IEEE Transactions on Circuits and Systems, 2024, all Open Access; Green Accepted Open Access; Green Open Access.

[161] Q. Yan, H. Wang, and S. Yang, "A learning-assisted bi-population evolutionary algorithm for distributed flexible job-shop scheduling with maintenance decisions," IEEE Transactions on Evolutionary Computation, 2025.

[162] Y. Zhang, T. Qu, M. Li, Z. Zhang, K. Zhang, Z. Hong, G. Q. Huang, and Z. Chen, "Digital twin-based multi-granularity synchronisation for production-warehousing under batch processing mode," International Journal of Production Research, 2024.

[163] H. Golpîra, "Smart energy-aware manufacturing plant scheduling under uncertainty: A risk-based multi-objective robust optimization approach," Energy, vol. 209, p. 118385, 2020.

[164] J. Huang, L. Gao, and X. Li, "A hierarchical multi-action deep reinforcement learning method for dynamic distributed job-shop scheduling problem with job arrivals," IEEE Transactions on Automation Science and Engineering, 2025.

[165] S. Yang, J. Wang, and Z. Xu, "Real-time scheduling for distributed permutation flowshops with dynamic job arrivals using deep reinforcement learning," Advanced Engineering Informatics, 2022.

[166] Z. Zhang, X. Li, B. Qian, H. Jin, R. Hu, and J. Yang, "Multi-agent cooperative multi-network group framework for energy-efficient distributed fuzzy flexible job shop scheduling problem," Applied Soft Computing, 2025.

[167] Q. Yang, Z.-M. Chen, Z.-S. Wang, and M. Deveci, "A refined two-stage fuzzy hybrid decision making support framework for enhancing logistics and manufacturing integration," Advanced Engineering Informatics, vol. 68, p. 103780, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.aei.2025.103780

[168] Y. Wang, R. Zheng, C. Liang, and J. Shi, "Inventory routing problem of automotive parts considering time-varying demands: A machine learning enhanced branch-and-price approach," Transportation Research Part E: Logistics and Transportation Review, 2025.

[169] Z. Hong, T. Qu, Y. Zhang, Z. Zhang, and G. Q. Huang, "Cloud-fog-edge based computing architechture and a hierarchical decision approach for distributed synchronized manufacturing systems," Advanced Engineering Informatics, vol. 65, p. 103386, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.aei.2025.103386

[170] Z. Zhang, T. Qu, G. Q. Huang, Y. Zhang, K. Zhao, and J. Zhang, "Towards a trusted synchronized decision-making method for social production logistics systems based on blockchain and digital twin," Robotics and Computer-Integrated Manufacturing, vol. 94, p. 102965, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.rcim.2025.102965

[171] Y. Zhang, Z. Hong, Y. Wei, T. Qu, and G. Q. Huang, "A digital twin and big data-driven opti-state control framework for production logistics synchronisation system," IET Collaborative Intelligent Manufacturing, vol. 7, no. 1, 2025. [Online]. Available: http://dx.doi.org/10.1049/cim2.70024

[172] Y. Zhang, T. Qu, Z. Hong, Z. Zhang, K. Zhang, M. Li, and G. Q. Huang, "Digital twin-based adaptive opti-state control approach for production-logistics synchronisation system," International Journal of Production Research, vol. 63, no. 24, pp. 10 208–10 240, 2025. [Online]. Available: http://dx.doi.org/10.1080/00207543.2025.2546030

[173] M. Breque, L. De Nul, and A. Petridis, "Industry 5.0: Towards a sustainable, human-centric and resilient european industry," Directorate-General for Research and Innovation, European Commission, Tech. Rep., 2021, catalogue number: KI-BD-20-021-EN-N.

[174] I. Gödri, "Improving delivery performance in high-mix low-volume manufacturing by model-based and data-driven methods," Applied Sciences (Switzerland), 2022, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[175] D. Kozjek, R. Vrabič, B. Rihtaršič, and P. Butala, "Big data analytics for operations management in engineer-to-order manufacturing," 2018, all Open Access; Gold Open Access.

[176] M. D. Sherwin, H. R. Medal, C. A. MacKenzie, and M. Hradecny, "A machine learning approach for engineer-to-order firms to predict supplier performance in critical supply chains," International Journal of Management Science and Engineering Management, 2025.

[177] J. Patalas-Maliszewska, G. Pajak, and K. Kowalczewska, "Ai-driven data analytics for enhanced mass customization in production," 2025, all Open Access; Gold Open Access.

[178] X. Gong, S. Wang, R. R. Jiao, and N. Z. Gebraeel, "Collaborative contracting for manufacturing-as-a-service (maas) by information content measurement and decision tree learning," Advanced Engineering Informatics, 2023.

[179] J. A. Afari, A. Gosavi, J. Hu, and R. J. Marley, "Data-driven layout design for smart remanufacturing: a flexible optimization model and a case study," International Journal on Interactive Design and Manufacturing (IJIDeM), vol. 19, no. 12, p. 8625–8639, 2025. [Online]. Available: http://dx.doi.org/10.1007/s12008-025-02296-5

[180] M. Vacchi, D. Settembre-Blundo, L. Iattici, A. M. Ferrari, R. Rosa, and N. Berselli, "From black box to analytical insight: A data-driven evaluation of technological sustainability in manufacturing supply chains," Supply Chain Analytics, vol. 12, p. 100171, 2025. [Online]. Available: http://dx.doi.org/10.1016/j.sca.2025.100171

[181] H. Wang and K. Zhu, "Collaborative optimization of intelligent manufacturing system and industrial design based on improved genetic algorithm," International Journal on Interactive Design and Manufacturing (IJIDeM), vol. 19, no. 12, pp. 8813–8824, 2025. [Online]. Available: http://dx.doi.org/10.1007/s12008-025-02316-4

[182] A. Cimino, F. Longo, G. Mirabelli, V. Solina, and P. Veltri, "Enhancing internal supply chain management in manufacturing through a simulation-based digital twin platform," Computers and Industrial Engineering, 2024, all Open Access; Hybrid Gold Open Access.

[183] T. Xia, G. Si, D. Wang, E. Pan, and L. Xi, "Progressive opportunistic maintenance policies for service-outsourcing network with prognostic updating and dynamical optimization," IEEE Transactions on Reliability, 2022.

[184] Z. Zhao and Q. Yuan, "Integrated scheduling of the production and maintenance of parallel machine job-shop considering stochastic machine breakdowns," 2022, all Open Access; Hybrid Gold Open Access.

[185] H. Jiang, Y. Li, X. Liu, G. Wang, and X. Zhang, "Solving extended flexible job shop scheduling problems with deep reinforcement learning," 2025, all Open Access; Gold Open Access.

[186] Y. Hu, Z. Zhang, J. Chen, D. Tang, and Q. Cai, "Graph-based deep reinforcement learning for dynamic scheduling of flexible job-shop considering worker fatigue and multi-skill factors," Applied Soft Computing, 2025.

[187] M. Yuan, Q. Yu, L. Zhang, S. Lu, Z. Li, and F. Pei, "Deep reinforcement learning based proximal policy optimization algorithm for dynamic job shop scheduling," Computers and Operations Research, 2025.

[188] A. Lorenzi, S. Genetti, C. C. Migliore Rambaldi, M. Roveri, and G. Iacca, "A hybrid constrained programming with genetic algorithm for the job shop scheduling problem," 2025, all Open Access; Hybrid Gold Open Access.

[189] A. Mombelli and A. Quilliot, "Synchronizing production planning and job scheduling: Milp models and exact algorithms," Journal of Combinatorial Optimization, 2025.

[190] H. Oukassi, A. Jaoua, S. Yacout, E. Negri, and M. Jaoua, "Towards standardizing the integration of digital twins in manufacturing systems," in Proceedings of the International Winter Simulation Conference, 2024, pp. 2939–2950.

[191] J. Pan, R. Zhong, B. Hu, Y. Feng, Z. Zhang, and J. Tan, "Smart scheduling of hanging workshop via digital twin and deep reinforcement learning," Flexible Services and Manufacturing Journal, 2025.

[192] Y. Du and L. Zhou, "Discrete manufacturing flexible transition: Research on digital twin of high-end equipment and intelligent algorithm closed-loop system," 2025.

[193] J. Geng and Q. Nie, "Probing an autonomous decision-making production mechanism of the digital twin shop-floor based on the industrial large model," 2025.

[194] G. Li, L. Zhang, and X. Zhao, "Discrete event simulation-driven method solving permutation flowshop scheduling problem in digital twins," IEEE Access, 2025, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[195] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," IEEE Access, vol. 8, pp. 108 952–108 971, 2020.

[196] D. Jones, C. Snider, A. Nassehi, J. Yon, and B. Hicks, "Characterising the digital twin: A systematic literature review," CIRP Journal of Manufacturing Science and Technology, vol. 29, pp. 36–52, 2020.

[197] J.-F. Yao, Y. Yang, X.-C. Wang, and X.-P. Zhang, "Systematic review of digital twin technology and applications," Visual Computing for Industry, Biomedicine, and Art, vol. 6, no. 1, p. 10, 2023.

[198] A. A. García-León and W. F. Torres Tapia, "Integrating production and preventive maintenance in the flexible job shop scheduling problem with setup times and sequence-dependent." 2025, all Open Access; Gold Open Access.

[199] A. Maierhofer, S. Trojahn, and F. Ryll, "Dynamic scheduling and preventive maintenance in small-batch production: A flexible control approach for maximising machine reliability and minimising delays," Applied Sciences (Switzerland), 2025, all Open Access; Gold Open Access.

[200] N. Ouahabi, A. Chebak, O. Kamach, and M. Zegrari, "Dynamic production scheduling and maintenance planning under opportunistic grouping," Computers and Industrial Engineering, 2025, all Open Access; Hybrid Gold Open Access.

[201] R. Zhong, Y. Feng, B. Hu, G. Zheng, X. Li, X. Chen, and J. Tan, "An assembly quality inspection framework for small-batch products via hcps and haas?" 2025, all Open Access; Gold Open Access.

[202] Y. Hillali, M. Zegrari, N. Alfathi, and C. Samir, "New approach for detecting variability in industrial assembly line balancing based on multi-criteria analysis," Automation, 2025, all Open Access; Gold Open Access.

[203] K. Petersen, S. Vakkalanka, and L. Kuzniarz, "Guidelines for conducting systematic mapping studies in software engineering: An update," Information and Software Technology, vol. 64, pp. 1–18, 2015. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950584915000646

[204] Q. Yan, H. Wang, and F. Wu, "Digital twin-enabled dynamic scheduling with preventive maintenance using a double-layer q-learning algorithm," Computers and Operations Research, 2022.

[205] X. Zhang, Z. Li, S. Fu, Q. Xu, Z. Du, and G. Yuan, "Optimization scheduling of dynamic industrial systems based on reinforcement learning," Applied Sciences (Switzerland), 2025, all Open Access; Gold Open Access.

[206] W. Zhong, Y. Li, Y. Wu, and Y. Wang, "Flexible job shop scheduling based on learnable graph convolutional attention networks and reinforcement learning," Journal of the Franklin Institute, 2025.

[207] S. Zhai, M. G. Kandemir, and G. Reinhart, "Predictive maintenance integrated production scheduling by applying deep generative prognostics models: approach, formulation and solution," Production Engineering, 2022, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[208] C. Li, X. Zhao, J. Feng, M. Yang, X. Wang, H. Gao, and M. Zhang, "A flexible job shop scheduling method via a hybrid dual attention network and mamba approach," 2025.

[209] X. Wang, Z. Liang, P. Zhong, D. Li, H. Li, and M. Liu, "Multi-objective scheduling for green flexible assembly job-shop system via multi-agent deep reinforcement learning with game theory," IEEE Access, 2025, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[210] S. Yang, H. Guo, J. Huang, and K. Han, "A deep reinforcement learning based approach for dynamic job shop scheduling considering variable processing time," 2024.

[211] A. D. Workneh and M. Gmira, "Deep q network method for dynamic job shop scheduling problem," Lecture Notes in Networks and Systems, 2023.

[212] ——, "Learning to schedule (l2s): adaptive job shop scheduling using double deep q network," Smart Science, 2023.

[213] C. Lyu, C. Dong, Q. Xiong, Y. Chen, Q. Weng, and Z. Chen, "Pathfinder: Deep reinforcement learning-based scheduling for multi-robot systems in smart factories with mass customization," Computers, Materials and Continua, 2025, all Open Access; Gold Open Access.

[214] B. Han and J. Yang, "Research on adaptive job shop scheduling problems based on dueling double dqn," IEEE Access, 2020, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[215] S. Zhai, A. Riess, and G. Reinhart, "Formulation and solution for the predictive maintenance integrated job shop scheduling problem," 2019, all Open Access; Green Accepted Open Access; Green Open Access.

[216] B. M. Méndez-Hernández, E. D. Rodríguez-Bazan, Y. Martínez-Jiménez, P. J. Libin, and A. Nowé, "A multi-objective reinforcement learning algorithm for jssp," Lecture Notes in Computer Science, 2019.

[217] L. Hua, H. Liu, and Y. Pan, "Solving industrial chain job scheduling problems through a deep reinforcement learning method with decay strategy," Information Sciences, 2025.

[218] C. Waubert de Puiseau, F. Wolz, M. Montag, J. Peters, H. Tercan, and T. Meisen, "Applying decision transformers to enhance neural local search on the job shop scheduling problem," AI (Switzerland), 2025, all Open Access; Gold Open Access.

[219] S. Lassoued and A. Schwung, "Introducing petrirl: An innovative framework for jssp resolution integrating petri nets and event-based reinforcement learning," Journal of Manufacturing Systems, 2024, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[220] A. K. Elsayed, E. K. ElSayed, and K. A. El-Dahshan, "Deep reinforcement learning based actor-critic framework for decision-making actions in production scheduling," 2021.

[221] D. Sun and H. Zang, "Research on the application of evolutionary algorithms enhanced by deep learning in intelligent manufacturing scheduling," 2024.

[222] J. Yan, T. Zhao, T. Zhang, H. Y. Chu, C. Yang, and Y. Zhang, "A dynamic scheduling method combining iterative optimization and deep reinforcement learning to solve sudden disturbance events in a flexible manufacturing process," Mathematics, 2025, all Open Access; Gold Open Access.

[223] L. Ding, D. Luo, M. Rauf, and L. Yue, "Deep reinforcement learning for data-driven scheduling in multi-variety and small-batch flexible job shops: Integrating fluid models for enhanced optimization," Computers and Industrial Engineering, 2025.

[224] G. Bencheikh, A. Letouzey, and X. Desforges, "An approach for joint scheduling of production and predictive maintenance activities," Journal of Manufacturing Systems, 2022, all Open Access; Green Accepted Open Access; Green Open Access.

[225] A. Bubak, B. Rolf, T. Reggelin, S. Lang, and H. Stuckenschmidt, "An lstm network-based genetic algorithm for integrated procurement and scheduling optimisation," International Journal of Production Research, 2025, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[226] P. Burggräf, J. Wagner, B. Koke, and M. Bamberg, "Performance assessment methodology for ai-supported decision-making in production management," 2020, all Open Access; Gold Open Access.

[227] A. Ghasemi, Y. T. Yeganeh, A. Matta, K. E. Kabak, and C. Heavey, "Deep learning enabling digital twin applications in production scheduling: Case of flexible job shop manufacturing environment," 2023.

[228] Y. He and K. E. Stecke, "Simultaneous part input sequencing and robot scheduling for mass customisation," International Journal of Production Research, 2022.

[229] J. He and J. Li, "Deep reinforcement learning based on graph neural network for flexible job shop scheduling problem with lot streaming," Lecture Notes in Computer Science, 2024.

[230] Y. Hu, J. Chen, Z. Zhang, D. Tang, and Q. Cai, "Dynamic scheduling for dual-resource constrained flexible job-shop via semantic-aware graph modelling and deep reinforcement learning," Knowledge-Based Systems, 2025.

[231] S. Lan, Y. Jiang, C. Yang, L. Wang, G. Q. Huang, W. Shen, and L. Zhu, "Knowledge guided drl for intelligent reconfiguration and scheduling in customized and personalized manufacturing workshop," IEEE Transactions on Industrial Informatics, vol. 22, no. 1, pp. 429–440, 2026. [Online]. Available: http://dx.doi.org/10.1109/TII.2025.3610442

[232] D. Lee, I. Park, and K. Kim, "A self-imitation learning approach for scheduling evaporation and encapsulation stages of oled display manufacturing systems," Robotics and Computer-Integrated Manufacturing, 2025.

[233] M. E. Leusin, E. M. Frazzon, M. Uriona-Maldonado, M. Kück, and M. M. Freitag, "Solving the job-shop scheduling problem in the industry 4.0 era," Technologies, 2018, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[234] R. Liu, R. Piplani, and C. Toro, "A deep multi-agent reinforcement learning approach to solve dynamic job shop scheduling problem," Computers and Operations Research, 2023.

[235] S. Lu, Y. Wang, M. Kong, W. Wang, W. Tan, and Y. Song, "A double deep q-network framework for a flexible job shop scheduling problem with dynamic job arrivals and urgent job insertions," Engineering Applications of Artificial Intelligence, 2024.

[236] G. E. Mejía, C. Montoya, S. Bolívar, and D. A. Rossit, "Job shop rescheduling with rework and reconditioning in industry 4.0: an event-driven approach," International Journal of Advanced Manufacturing Technology, 2022.

[237] E. Nejati, E. Ghaedy-Heidary, A. Ghasemi, and S. A. Torabi, "A machine learning-based simulation metamodeling method for dynamic scheduling in smart manufacturing systems," Computers and Industrial Engineering, 2024, all Open Access; Hybrid Gold Open Access.

[238] J. Peng, J. Wang, D. Wang, A. Kimmig, and J. G. Ovtcharova, "The research of flexible scheduling of workshop based on artificial fish swarm algorithm and knowledge mining," Lecture Notes in Computer Science, 2020, all Open Access; Green Accepted Open Access; Green Open Access.

[239] M. S. Pereira, P. T. Aquino-Jr, C. A. Mattos, and F. Lima, "Energy estimation and production scheduling in job shop using machine learning," IEEE Access, 2024, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[240] Y. Qu, J. Wang, B. Jiang, S. Cheng, Y. Wang, P. Wu, X. Ming, and X. Chu, "Self-decision mechanisms of smart production systems based on improved uncertainty theory and user-cfa," Computers and Industrial Engineering, vol. 183, p. 109453, 2023. [Online]. Available: http://dx.doi.org/10.1016/j.cie.2023.109453

[241] D. Rangel-Martínez and L. A. Ricardez–Sandoval, "A recurrent reinforcement learning strategy for optimal scheduling of partially observable job-shop and flow-shop batch chemical plants under uncertainty," Computers and Chemical Engineering, 2024, all Open Access; Hybrid Gold Open Access.

[242] E. Salatiello, S. Vespoli, G. Guizzi, and A. Grassi, "Long-sighted dispatching rules for manufacturing scheduling problem in industry 4.0 hybrid approach," Computers and Industrial Engineering, 2024, all Open Access; Hybrid Gold Open Access.

[243] J. Senoner, B. Kratzwald, M. Kuzmanovic, T. H. Netland, and S. Feuerriegel, "Addressing distributional shifts in operations management: The case of order fulfillment in customized production," Production and Operations Management, 2023, all Open Access; Green Accepted Open Access; Green Open Access; Hybrid Gold Open Access.

[244] J. C. Serrano-Ruiz, J. Mula, and R. Poler, "Job shop smart manufacturing scheduling by deep reinforcement learning," Journal of Industrial Information Integration, 2024, all Open Access; Hybrid Gold Open Access.

[245] J. Shi, F. Qiao, Y. Ma, J. Liu, and J. Wang, "Production-logistics collaborative scheduling in dynamic flexible job shops via multi-objective deep reinforcement learning," 2024.

[246] J. Shi, F. Qiao, J. Liu, Y. Ma, D. Wang, and C. Ding, "Production-logistics collaborative scheduling in dynamic flexible job shops using nested-hierarchical deep reinforcement learning," Advanced Engineering Informatics, 2025.

[247] D. Tremblet, S. Thevenin, and A. B. Dolgui, "Predicting makespan in flexible job shop scheduling problem using machine learning," 2022, all Open Access; Gold Open Access.

[248] J. Wan, X. Li, H. Dai, A. Kusiak, M. Martínez-García, and D. H. Li, "Artificial-intelligence-driven customized manufacturing factory: Key technologies, applications, and challenges," Proceedings of the IEEE, 2021, all Open Access; Green Accepted Open Access; Green Open Access.

[249] Z. Wang and W. Liao, "Job shop scheduling problem using proximal policy optimization," 2023.

[250] R. Wang, C. Liu, X. Wang, H. Shi, and S. Yang, "Experience replay-driven evolutionary reinforcement learning for dynamic flexible flow shop scheduling," 2025.

[251] Q. Xiao, K. Wang, C. Ma, and Y. Chen, "Material delivery optimization for make-to-order reconfigurable job shops using an improved chaotic multi-verse algorithm," Swarm and Evolutionary Computation, 2025, all Open Access; Hybrid Gold Open Access.

[252] F. Xu and J. Li, "A multi-action reinforcement learning framework via pointer graph neural network for flexible job-shop scheduling problems with resource transfer," Lecture Notes in Computer Science, 2024.

[253] B. Yan, X. Liu, S. Lu, C. Hu, X. Wang, and Z. Zhou, "Deep reinforcement learning approach for a dynamic flexible job shop problem with sequence dependent setup times," Computers and Industrial Engineering, 2025.

[254] J. Yang, H. Wang, and J. Xiong, "Evolutionary strategies with dual graph reinforcement learning for flexible job shop scheduling problem," 2025.

[255] J. C. Yepes-Borrero, J. Alcaraz, M. López-García, and M. Villaizán-Vallelado, "Unrelated parallel machine scheduling problem with setup times and additional resources: an enhanced metaheuristic to address resource-related infeasibilities," Annals of Operations Research, 2026. [Online]. Available: http://dx.doi.org/10.1007/s10479-026-07065-5

[256] W. Yi, Q. Deng, B. Wang, Y. Chen, and Z. Pei, "Dynamic flexible job shop scheduling problem considering multiple types of dynamic events," International Journal of Production Research, 2025.

[257] W. Yoo, J. Kim, D. M. Concha, and C. Lee, "Optimizing markov decision process state design for deep reinforcement learning manufacturing scheduling using bayesian optimization," Journal of Computational Design and Engineering, 2025, all Open Access; Gold Open Access.

[258] Y. Zeng, A. Liu, S. Huang, X. Chen, W. Liang, and X. Zhou, "Permutation-invariant and equivariant multiagent reinforcement learning for flexible manufacturing in industrial iot," IEEE Internet of Things Journal, 2025.

[259] Y. Zhang, H. Zhu, D. Tang, T. Zhou, and Y. Gui, "Dynamic job shop scheduling based on deep reinforcement learning for multi-agent manufacturing systems," Robotics and Computer-Integrated Manufacturing, 2022.

[260] M. Zhang, Y. Lu, Y. Hu, N. Amaitik, and Y. Xu, "Dynamic scheduling method for job-shop manufacturing systems by deep reinforcement learning with proximal policy optimization," Sustainability (Switzerland), 2022, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[261] B. Zhang, H. Sang, C. Lu, L. Meng, Y. Song, and X. Jiang, "Integrated heterogeneous graph and reinforcement learning enabled efficient scheduling for surface mount technology workshop," Information Sciences, 2025.

[262] L. Zheng, X. Chen, C. Zhuang, J. Liu, Y. Zhang, and L. Lai, "Dynamic scheduling for flexible job-shop with reconfigurable manufacturing cells considering dynamic job arrivals based on deep reinforcement learning," International Journal of Production Research, vol. 63, no. 20, pp. 7427–7459, 2025. [Online]. Available: http://dx.doi.org/10.1080/00207543.2025.2497961

[263] R. Zhong, Y. Feng, A. Zhang, C. Li, K. Jin, Z. Li, and J. Tan, "Solving flexible job-shop problem considering skilled workers via multi-domain graph attention network," International Journal of Production Research, 2025.

[264] T. Zhou, D. Tang, H. Zhu, and L. Wang, "Reinforcement learning with composite rewards for production scheduling in a smart factory," IEEE Access, 2021, all Open Access; Gold Open Access; Green Accepted Open Access; Green Open Access.

[265] X. Zhou, D.-k. Chae, and S.-c. Lee, "Practical algorithms for weakly flexible job scheduling for smart mold component process," Future Generation Computer Systems, 2024.